AETHERIX

Autonomous Extraterrestrial High-throughput Enhancing Routing and Inter-planetary eXchange

Building an Interplanetary Communication Network with DTN,
Quantum Communication, and Space-Based Infrastructure for Mars Mission Support

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matx104.github.io/AETHERIX | github.com/matx104/AETHERIX

Speaker Notes:

State your name clearly. Read the topic number and title exactly as on the exam paper. Pause to let examiners see it. Point to the logo. This is your first impression. (30 seconds)

PRESENTATION AGENDA

01 The Challenge
Why interplanetary communication is hard

02 AETHERIX Architecture
DTN + AI Routing + Quantum Security

03 DTN & Bundle Protocol v7
Store-and-forward networking foundation

04 5-Tier Network Topology
241 nodes across Earth and Mars

05 Optical Link Budget
1550 nm laser performance analysis

06 RL-Based Routing
Multi-agent federated Q-learning

07 Quantum Security (QKD)
BB84/E91 with repeater chains

08 Orbital Mechanics
Contact windows and synodic period

09 Mars Mission Scenario
End-to-end simulation walkthrough

10 Performance Comparison
AETHERIX vs current systems

11 Roadmap & Standards
CCSDS, IETF, deployment phases

12 Conclusion & Q&A
Summary and live demo

Speaker Notes:

Quick overview of what we will cover. 13 topics across 47 slides. About 20 minutes. (20 seconds)

WHAT IS AETHERIX?

Autonomous Extraterrestrial High-throughput Enhancing Routing and Inter-planetary eXchange

What is AETHERIX?

An architecture for delay-tolerant networking (DTN) between Earth and Mars, combining:

- Bundle Protocol v7 (BPv7) via ION-DTN
- Reinforcement Learning routing (replacing static CGR)
- Quantum Key Distribution (QKD) for security
- Hybrid optical/RF communications
- 5-tier topology with 241 nodes

The Problem

- Earth-Mars: 54.6M – 401M km distance
- One-way light delay: 3 – 22 minutes
- Solar conjunction: 2-week total blackout
- TCP/IP fundamentally cannot work
- Current systems: 0.5 – 6 Mbps only

AETHERIX delivers a complete interplanetary networking solution

combining DTN protocols, AI-driven routing, quantum-secure keys, and hybrid optical/RF links for Mars mission support.

10-100×	>95%	241	189
Faster	Availability	Nodes	Tests

Speaker Notes:

This slide sets up the narrative arc. First explain what AETHERIX is in plain language - it's like the postal service for interplanetary space. Then pivot to the problem: TCP/IP was never designed for space. 22-minute delays break every assumption. Solar conjunction blackouts. Static routing. Vulnerable crypto. Each problem maps to one of our solutions. (1.5 minutes)

THE DISTANCE

Why Space Communication is Fundamentally Different

Earth → Mars	Distance	Light Time	Data Rate
Perihelion (closest)	54.6M km	~3 min	100-200 Mbps
Average	225M km	~12.5 min	10-20 Mbps
Aphelion (farthest)	401M km	~22 min	2-5 Mbps

TCP/IP Assumption	Space Reality	Impact
RTT < 1 second	6 – 44 min RTT	Timeout failure
Always connected	Scheduled contacts	Connection drops
End-to-end path	No persistent path	Routing impossible
Fast acknowledgments	ACKs take minutes	Window collapse
Low packet loss	High loss (optical)	Retransmit storms

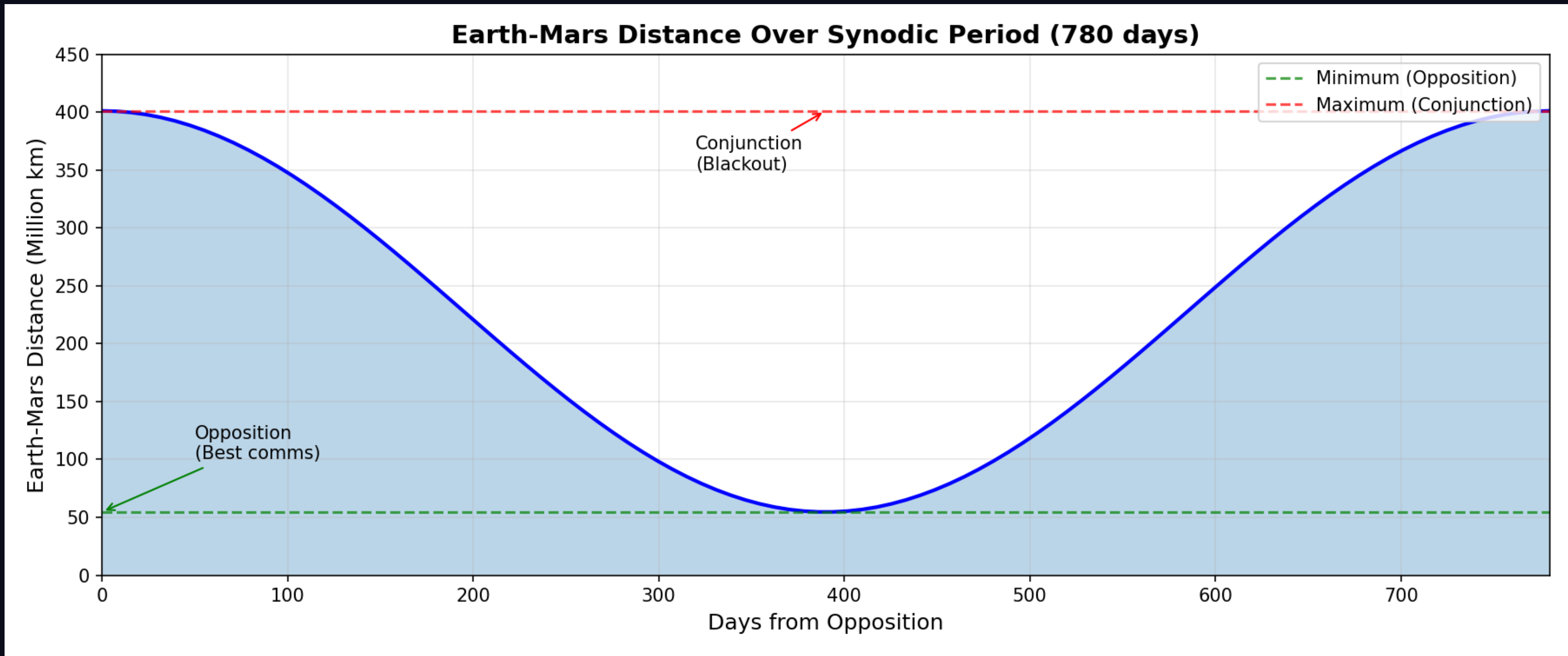
"At aphelion, a single ping-pong takes 44 minutes.
No TCP session can survive that. We need an entirely different networking paradigm."

Speaker Notes:

Start with the scale. 54.6M to 401M km. Light itself takes 3-22 minutes one way. TCP/IP was designed for sub-second round trips. In space, by the time a packet acknowledgment returns, the link may be gone. Solar conjunction causes 2-week blackout. This is why NASA calls it Delay-Tolerant Networking. (1.5 minutes)

EARTH-MARS DISTANCE OVER TIME

780-Day Synodic Period



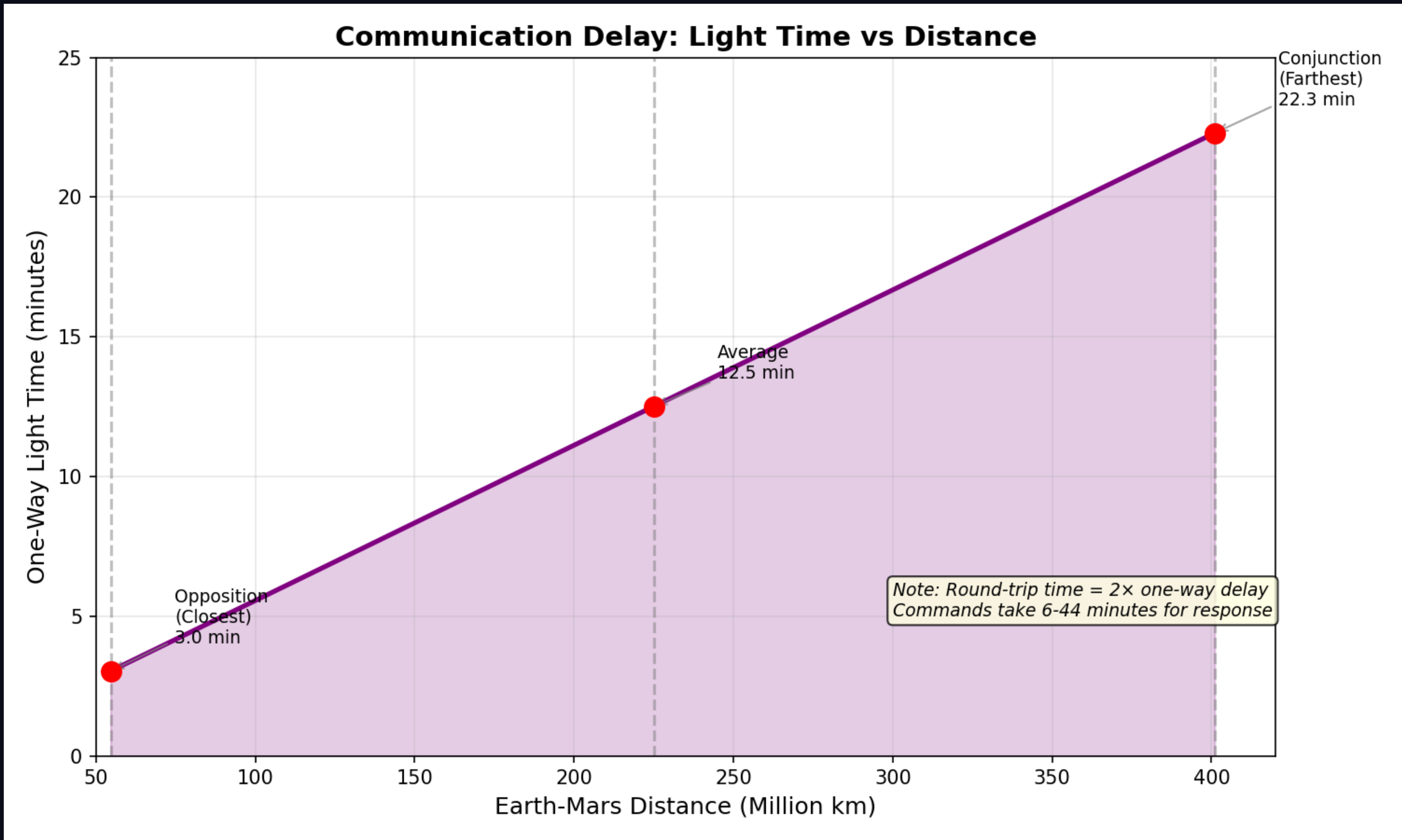
Distance varies 7× from 54.6M to 401M km · Solar conjunction causes ~14-day blackout every 26 months

Speaker Notes:

The distance chart shows the fundamental challenge: a 7x variation over the synodic period. At opposition, 55 million km. At conjunction, over 400 million km with the Sun blocking direct communication.

ONE-WAY LIGHT-TIME DELAY

3–22 Minutes Depending on Distance



Light-time delay drives every protocol design decision in AETHERIX

Speaker Notes:

Light-time delay ranges from 3 minutes at closest approach to 22 minutes at maximum distance. TCP/IP expects sub-second round trips — this is why we need DTN.

THE ANSWER: DELAY-TOLERANT NETWORKING

Decouple Communication from Connectivity

CREATE	STORE	WAIT	FORWARD	DELIVER
Bundle	Locally	For Link	Next Hop	Destination

TCP/IP: END-TO-END CONNECTION (FAILS IN SPACE)

Source - - - Router - - - Router - - - Dest

DTN: STORE-AND-FORWARD (WORKS IN SPACE)

SOURCE
Create Bundle
Priority: P2
CUSTODY: Source

link up →

NODE A
Forward bundle
Custody accepted
STORE: 2 bundles

no link - - -

NODE B
Storing bundle
Waiting for link...
WAIT 14 min

link up →

NODE C
Forward bundle
Custody accepted
CUSTODY: Node C

→

DEST
Delivered

Key: Custody Transfer
Each node takes responsibility for the bundle

Data Never Lost
Bundle stored until next hop confirms

No End-to-End Needed
Hop-by-hop, works with 22-min delays

Three Convergence Layers
LTP (RFC 5326) — Deep space
TCPCL (RFC 7242) — Earth segment
UDP-CL — Optical ISL

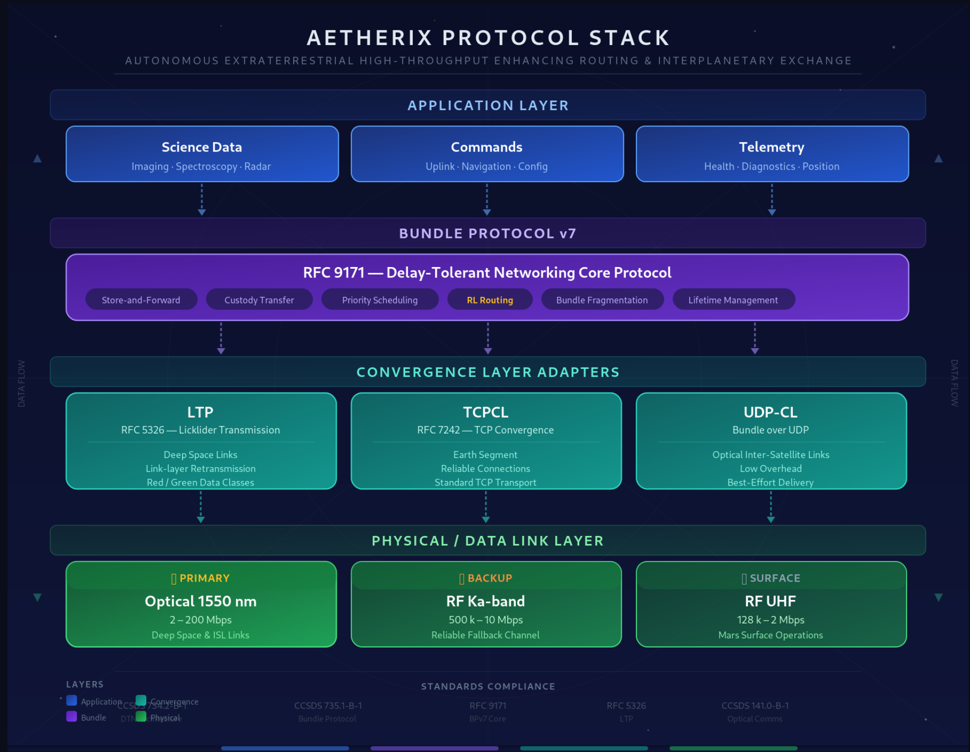
Speaker Notes:

The key insight: instead of requiring an end-to-end connection like TCP, DTN works like the postal service. Each node takes custody of your data and forwards it when a link becomes available. Three pillars: BPv7 for the protocol, RL for intelligent routing, QKD for security. (1.5 minutes)

SYSTEM ARCHITECTURE

5 Integrated Modules for Interplanetary Communication

Infrastructure	Link budget calculations, optical/RF analysis
Routing	RL agents, BPv7 bundles, forwarding engine
Security	BB84/E91 QKD, repeater chains, privacy amplification
Orbital	Contact windows, Doppler, celestial body database
Simulation	End-to-end scenario engine, policy routing



Layer	Module	Engine	Showcase
Infrastructure	link_budget.py	OpticalLinkBudget	Earth-Mars scenarios
Routing	rl_agent.py	RLRoutingAgent	Federated Q-learning
Security	qkd.py	BB84/E91	Quantum key exchange
Orbital	contact_windows.py	ContactPredictor	Synodic period
Simulation	simulator.py	SimEngine	Full mission sim

Speaker Notes:

Show the architecture. Six core modules feed into the simulation engine, which feeds the web showcase. Standards compliance at the bottom. Point to each module as you explain. (1 minute)

SYSTEM ARCHITECTURE DIAGRAM

AETHERIX PLATFORM

INFRASTRUCTURE
link_budget.py
rf_link_budget.py

FSPL, EIRP, margin
Ka/X/S/UHF bands
CCSDS 141.0-B-1

ROUTING
rl_agent.py
bundle.py / ltp.py

Q-learning, federated
BPv7, LTP, TCPCL, UDP-CL
RFC 9171, RFC 5326

SECURITY
qkd.py
repeater_chain.py

BB84, E91 protocols
CASCADE reconciliation
NIST FIPS 203/204 (PQC)

ORBITAL
contact_windows.py
topology.py / doppler.py

5-tier, 241 nodes
Synodic period modeling
Doppler compensation

SIMULATION ENGINE
simulator.py · policy_engine.py · training.py · multi_agent.py · forwarding_engine.py

WEB SHOWCASE — 10 Interactive Demos
matx104.github.io/AETHERIX · 149 tests · 27 Python modules

CCSDS 734.2-B-1 · CCSDS 735.1-B-1 · CCSDS 141.0-B-1

RFC 9171 · RFC 5326 · RFC 7242 · NIST FIPS 203/204

Speaker Notes:

Architecture diagram showing source modules feeding simulation engine and web demos.

BUNDLE PROTOCOL v7 — DEEP DIVE

RFC 9171 — The Foundation of Interplanetary Networking

APPLICATION LAYER

Science data, commands, imagery

BUNDLE PROTOCOL v7 (RFC 9171)

Store-and-Forward | Custody Transfer | Priority (P0–P4)

CONVERGENCE LAYERS

LTP (deep space) | TCPCL (Earth) | UDP-CL (optical ISL)

PHYSICAL LAYER

Optical 1550nm (2–200 Mbps) | RF Ka-band | UHF (surface)

Priority	Name	Delay Target	Example
P0	Emergency	< 1 min	Anomaly alerts
P1	Expedited	< 30 min	Solar storm warnings
P2	Standard	< 24 hrs	Science data
P3	Normal	< 7 days	Telemetry
P4	Bulk	< 30 days	Software updates

HOW IT WORKS

1. Source creates bundle (payload + metadata + lifetime)
2. Forward to next hop when link becomes available
3. Store locally during outages and disruptions
4. Custody transfer — next node accepts responsibility
5. Repeat hop-by-hop until destination reached

STANDARDS COMPLIANCE

RFC 9171 — Bundle Protocol Version 7

RFC 4838 — DTN Architecture

RFC 5326 — Licklider Transmission Protocol

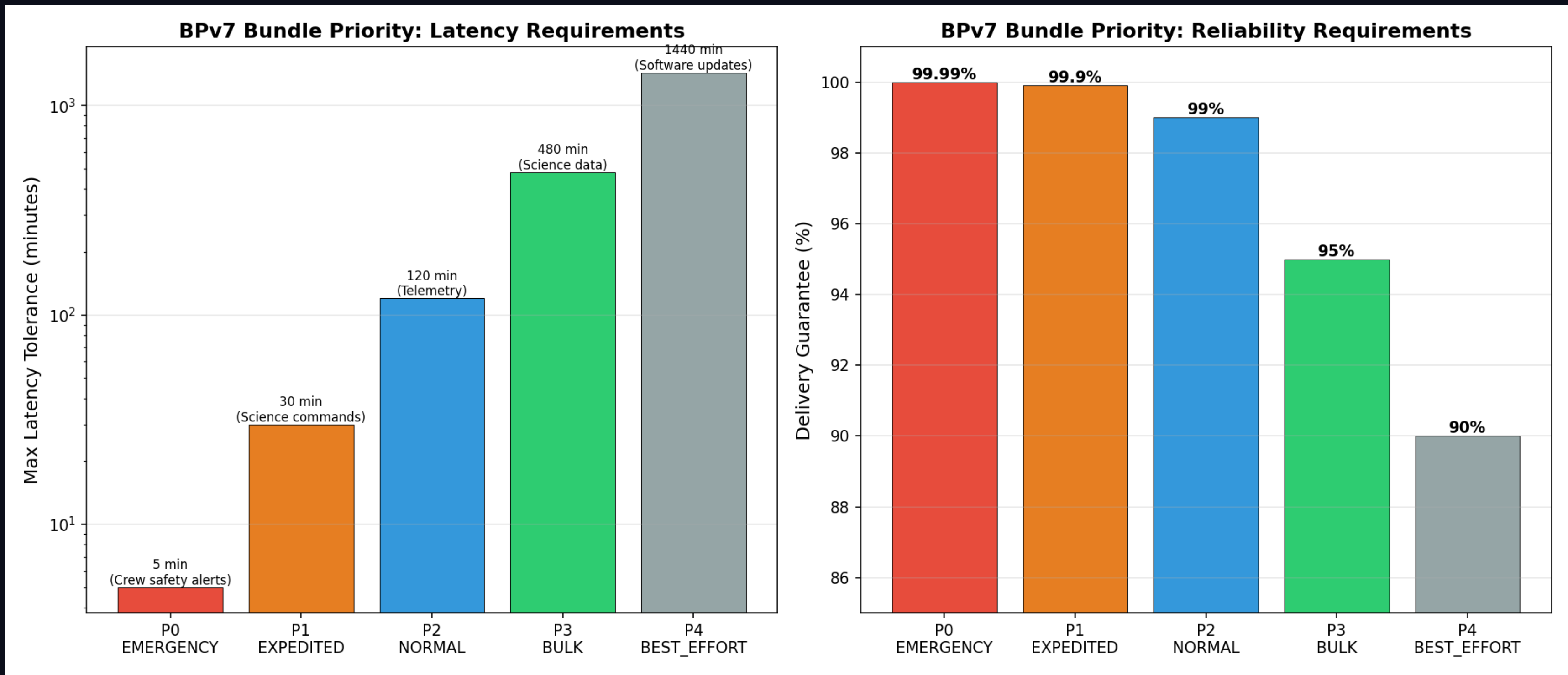
RFC 9172 — Bundle Protocol Security (BPSec)

CCSDS 734.2-B-1 — CCSDS Bundle Protocol

CCSDS 734.3-B-1 — Schedule-Aware Bundle Routing

BUNDLE PRIORITY CLASSES

Bandwidth Allocation Across Priority Levels



P0 Emergency through P4 Bulk · Deadline-aware scheduling · Preemption for critical traffic

Speaker Notes:

Priority class distribution showing how bandwidth is allocated. Emergency traffic preempts everything. The deadline-aware scheduler ensures no bandwidth is wasted.

DTN STORE-AND-FORWARD

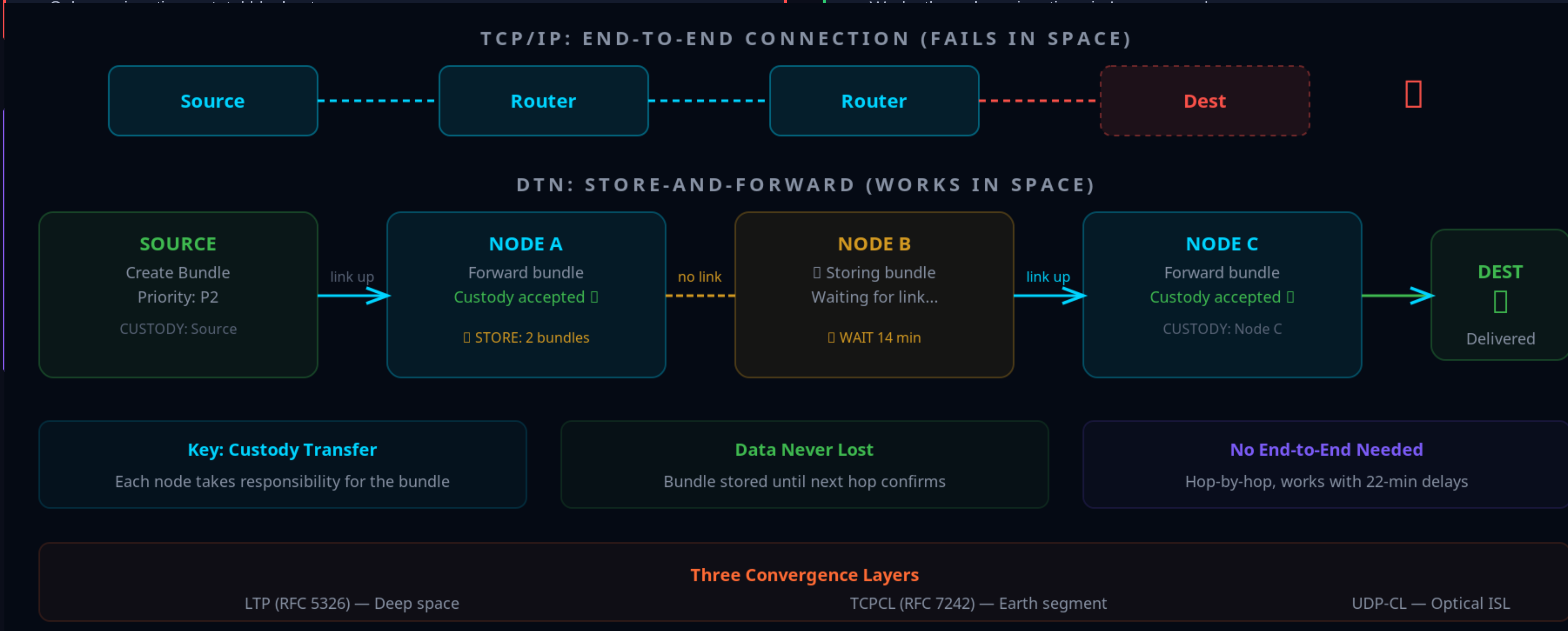
How Bundles Traverse the Interplanetary Network

TCP/IP FAILS

- Timeout at 44 min RTT → connection reset
- No persistent path → routing impossible
- ACKs take minutes → window collapse
- High BER → retransmission storms

DTN WORKS

- No end-to-end connection required
- Store locally → wait for contact window
- Custody transfer: hop-by-hop reliability
- Priority classes: P0–P4 for urgency



Speaker Notes:

Walk through the store-and-forward process. Bundle arrives, gets stored, node waits for next contact opportunity, then forwards. If link drops, bundle stays stored and retries. No data loss. This is fundamentally different from TCP's end-to-end retransmission. (1.5 minutes)

5-TIER NETWORK TOPOLOGY

241 Nodes — No Single Point of Failure

TIER 1: Earth Ground (6 nodes)

DSN Goldstone, Madrid, Canberra + Mission Control

TIER 2: Earth Orbital (51 nodes)

3 GEO relays + 48 LEO laser constellation

TIER 3: Deep Space (4 nodes)

ES-L4, ES-L5 Lagrange relays + Transfer orbit

TIER 4: Mars Orbital (4 nodes)

2 Areostationary + 2 polar orbiters

TIER 5: Mars Surface (176 nodes)

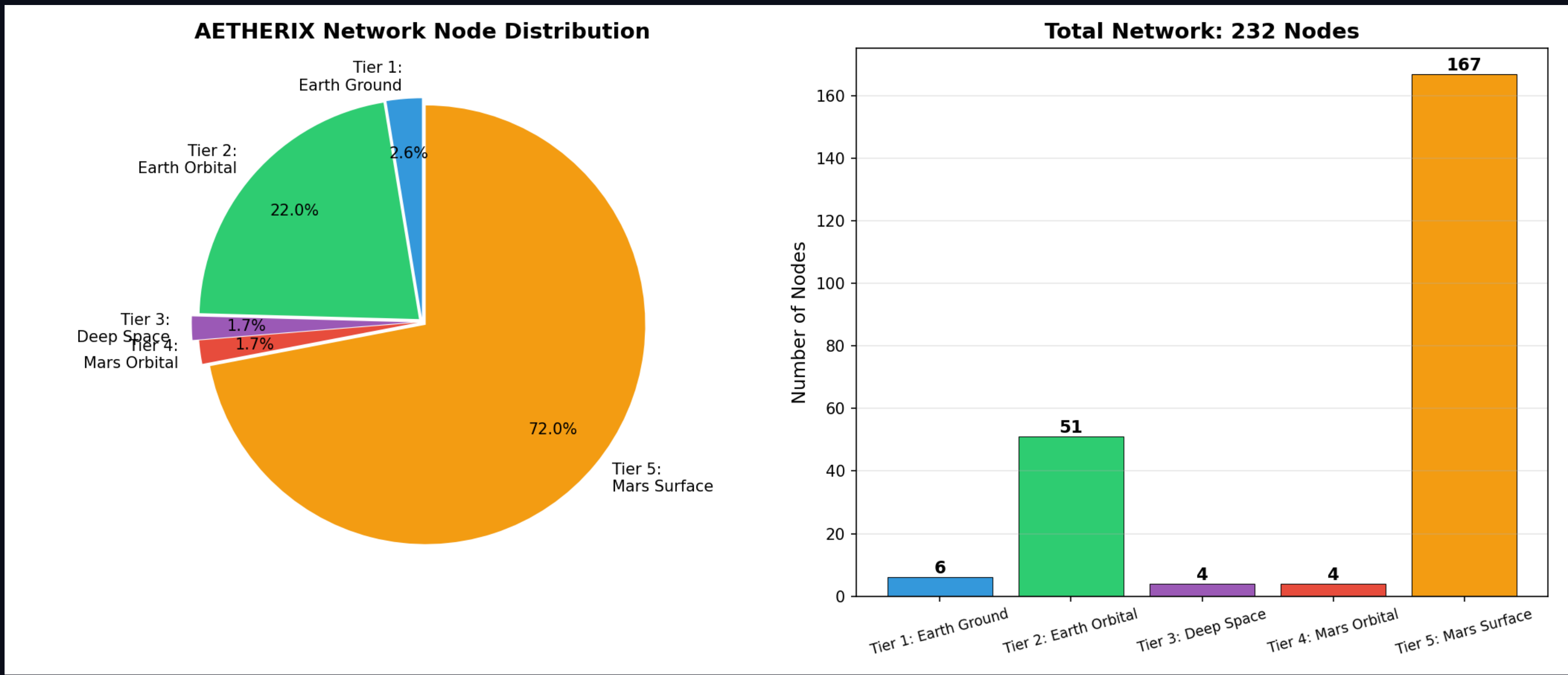
Bases, rovers, drones, sensor network

DESIGN PHILOSOPHY

Multiple redundant paths at every tier ensure no single point of failure. Lagrange point relays provide coverage during solar conjunction blackouts. Quantum repeaters co-located at L4/L5 enable end-to-end key distribution.

NODE DISTRIBUTION ACROSS 5 TIERS

241 Nodes Total



Mars Surface (167) is the largest tier · Deep Space (4) provides critical Lagrange relay coverage

Speaker Notes:

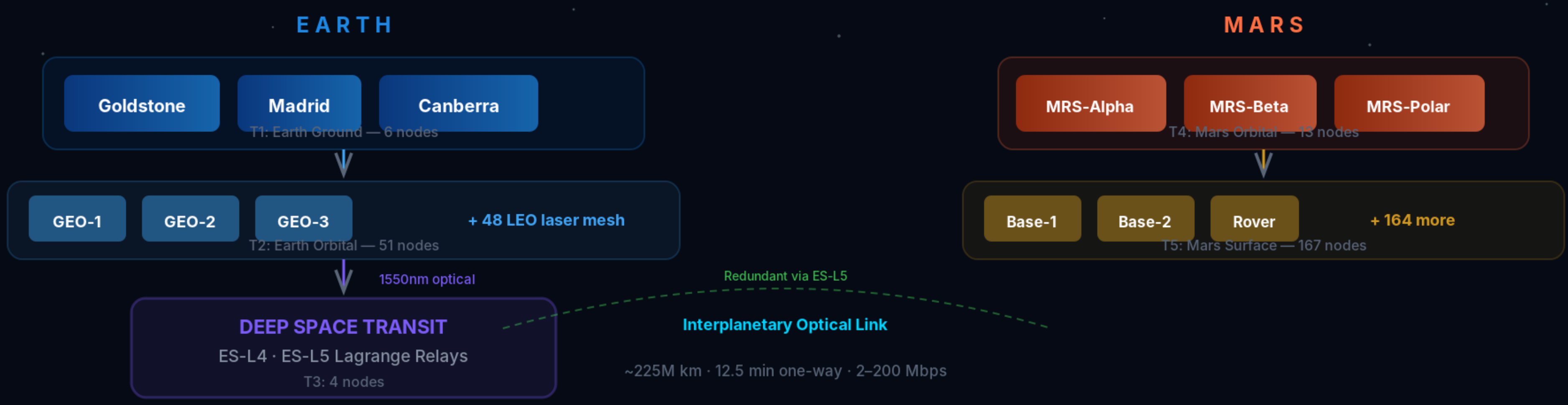
The tier distribution shows where the 241 nodes sit. Mars Surface dominates with 167 nodes. The 4 deep space nodes at Lagrange points are few but critical for conjunction survival.

Speaker Notes:

Visual overview of the 5-tier topology with 3 redundant paths.

5-TIER NETWORK DIAGRAM

241 Nodes from Earth to Mars Surface



241	5	3	0
Total Nodes	Network Tiers	Redundant Paths	Single Points of Failure

LINK CHARACTERISTICS

T1→T2	**DSN uplink**	Ka-band · GEO relay	LEO ISL	**Optical mesh**	10 Gbps · 48 satellites
T2→T3	**1550nm optical**	100 Mbps · Laser uplink	Latency	**3–22 min one-way**	Distance dependent
T3→T4	**Deep space laser**	2–200 Mbps · 225M km	Conjunction	**50–70% via Lagrange**	No total blackout
T4→T5	**UHF/X-band**	2 Mbps · Surface downlink	Security	**QKD + PQC end-to-end**	BB84 · ML-KEM

Speaker Notes:

Network topology visualization.

OPTICAL COMMUNICATIONS

1550 nm Laser — 10–100× Faster Than RF

200 Mbps

Peak Data Rate

At perihelion (54.6M km)

1550 nm

Wavelength

Telecom heritage lasers

33×

vs Current RF

MRO peak: 6 Mbps

KEY EQUATIONS

$FSPL = (4\pi d/\lambda)^2$ Free-Space Path Loss

$G = (\pi D/\lambda)^2$ Antenna Gain

$Pr = Pt \cdot Gt \cdot Gr / FSPL$ Received Power

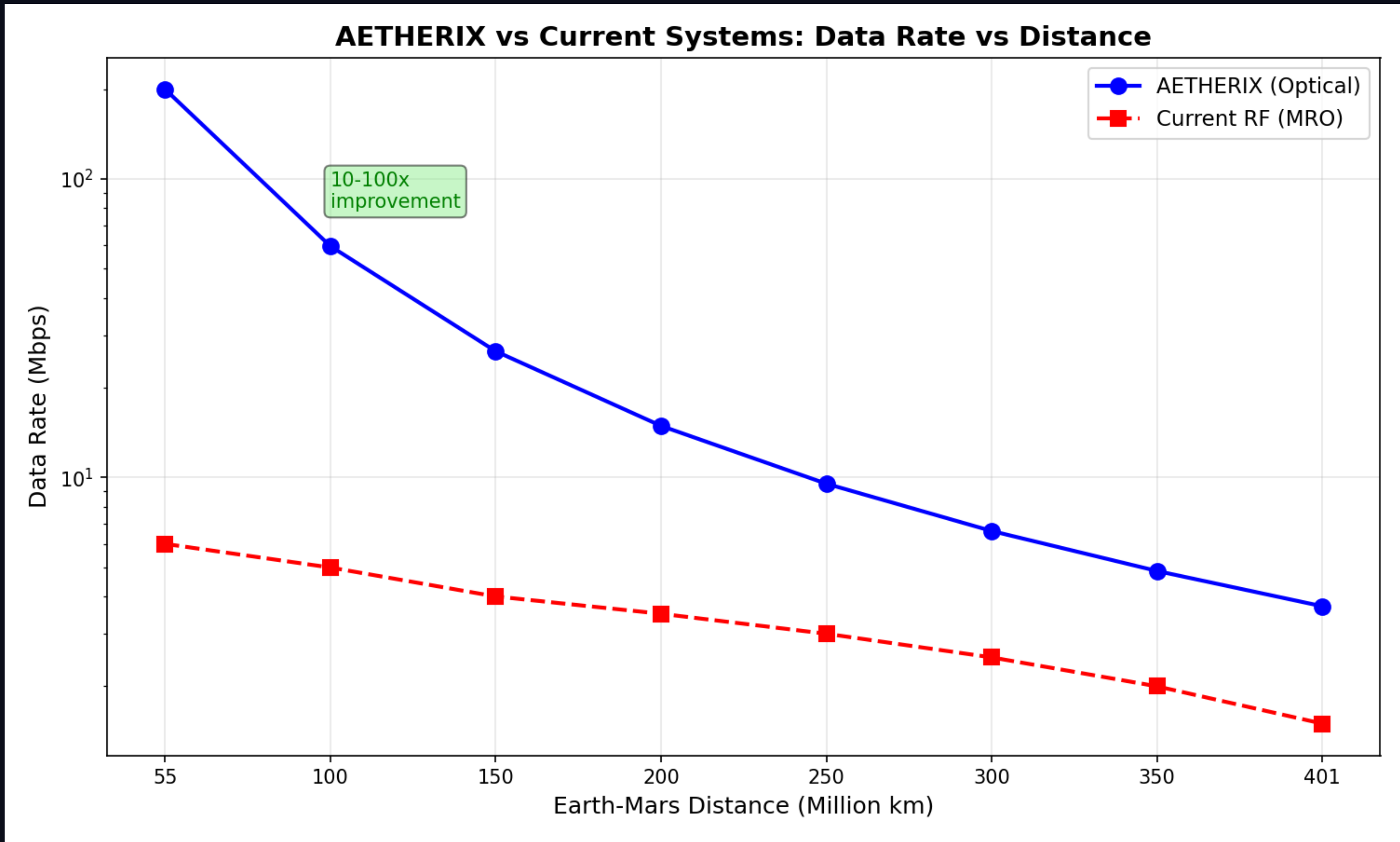
Parameter	Value	Rationale
TX Power	5 W (37 dBm)	Space-qualified laser
TX Aperture	22 cm	Mars orbiter class
RX Aperture	1.0 m	DSN-scale telescope
Wavelength	1550 nm	Telecom heritage
Efficiency	0.55	Conservative estimate

Speaker Notes:

RUN THE LIVE DEMO from the Link Budget page. Show the 3 distance scenarios. 1550nm was chosen for telecom heritage and eye safety. FSPL at average distance is -365 dB. The telescope apertures are realistic for spacecraft. RF backup for reliability. (2 minutes)

DATA RATE VS DISTANCE

1550 nm Optical Link Performance



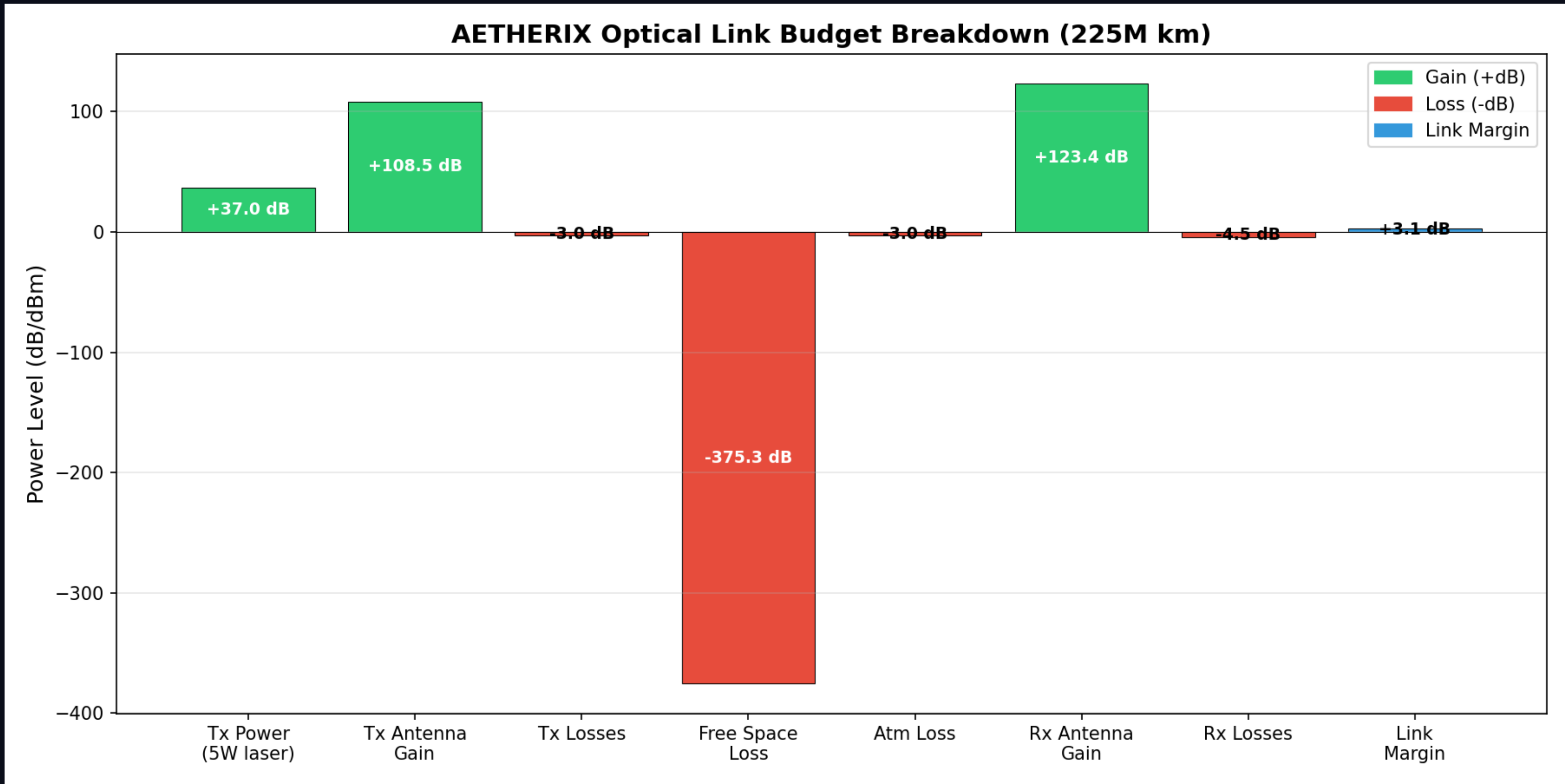
200 Mbps at closest approach to 2 Mbps at maximum distance · 10–100× improvement over RF

Speaker Notes:

Data rate degrades from 200 Mbps at closest approach to 2 Mbps at maximum distance — but even minimum is competitive with current RF systems.

OPTICAL LINK BUDGET BREAKDOWN

Where the Decibels Go



FSPL dominates at ~280 to ~310 dB · Optical aperture gain recovers >100 dB

Speaker Notes:

Link budget breakdown showing where the decibels go — free-space path loss is the dominant factor, compensated by high-gain optical apertures.

EARTH–MARS DATA JOURNEY

7 Hops from Perseverance Rover to Mission Control

Hop	From	To	Link	Time
1	Perseverance	MRS-Alpha	UHF	~5s
2	MRS-Alpha	MRS-Polar	Optical ISL	~1s

HOP DETAIL

HOP 1
Rover□UHF
UHF 400 MHz
~400 km · 2 Mbps

HOP 2
UHF□Areostat
17,032 km alt
256 kbps

HOP 3
Areostat□Polar
Optical ISL
10 Gbps

HOP 4-5
Deep Space
1550nm laser
225M km · 10-20 Mbps

HOP 6
LEO Mesh
Optical ISL
10 Gbps

HOP 7
DSN□JPL
Ka-band
DELIVERED □

Total Transit: ~13 min
vs 12.5 min light-time!

DTN Overhead: <5%
Store-and-forward adds minimal latency

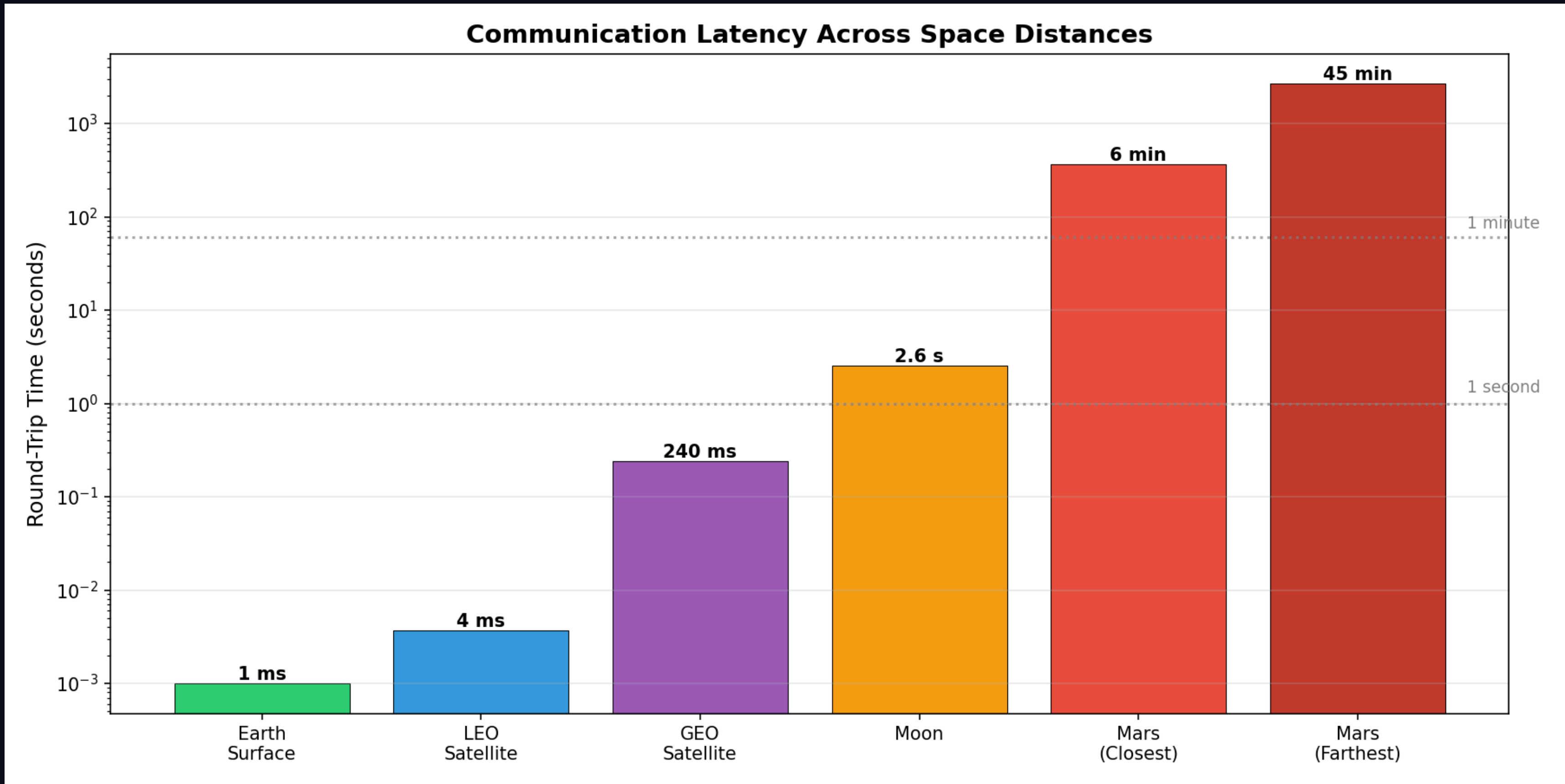
QKD Secured
End-to-end quantum encryption

□ If any link drops mid-transfer, the bundle is stored safely — zero data loss

Speaker Notes:
Interactive journey visualization.

LATENCY: TCP vs DTN vs AETHERIX

Protocol Overhead Comparison



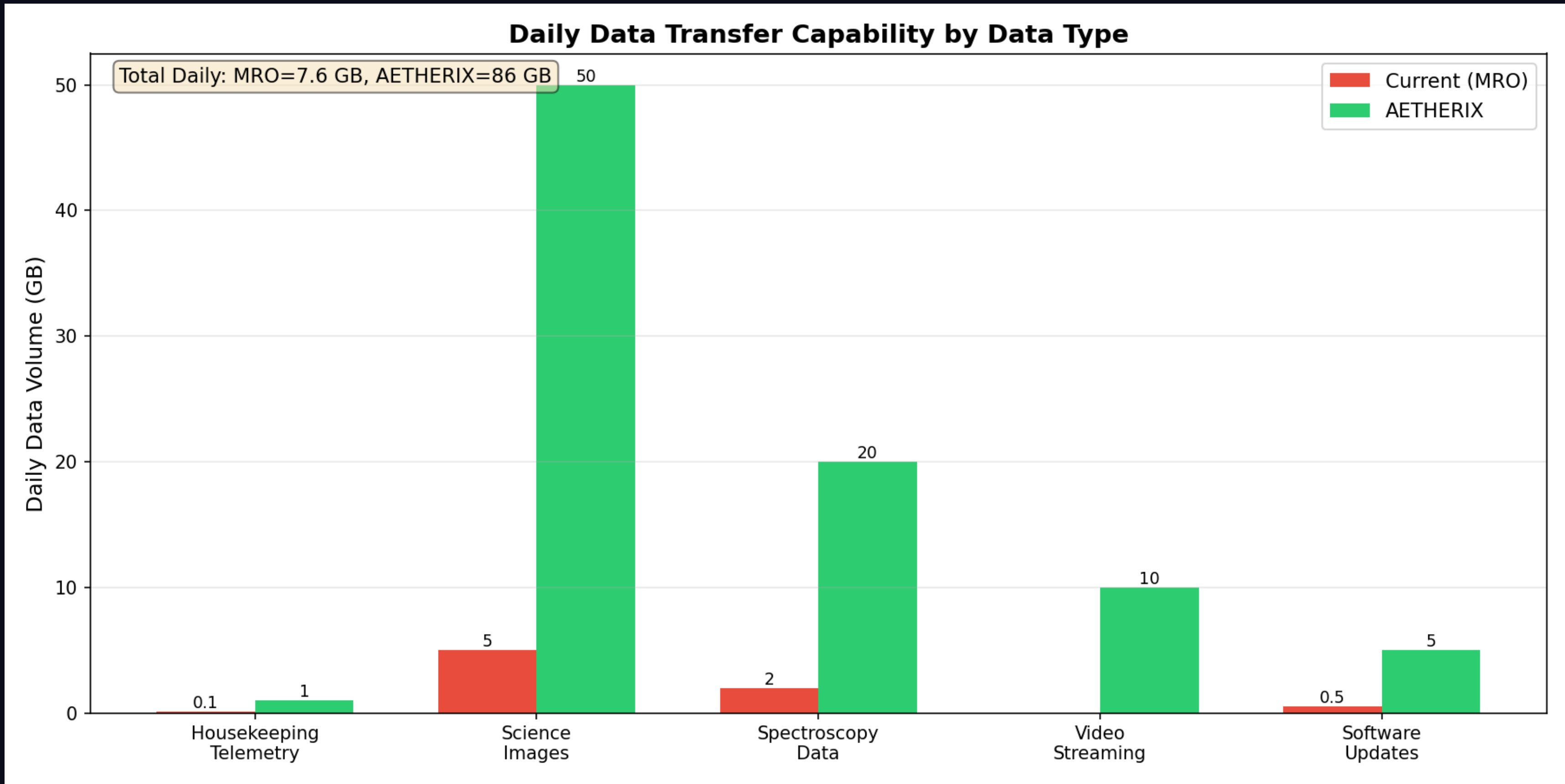
DTN adds <5% overhead beyond physical light-time · TCP fails catastrophically at interplanetary distances

Speaker Notes:

Latency comparison showing TCP failing catastrophically, while DTN adds under 5% overhead beyond the physical light-time limit.

DAILY DATA VOLUME COMPARISON

AETHERIX vs Current Systems



10–20x daily data volume improvement over current Mars missions

Speaker Notes:

AETHERIX delivers 10 to 20 times more data per day than current Mars missions.

RL-BASED ROUTING — AI INNOVATION

Multi-Agent Federated Q-Learning Replaces Static CGR

CGR LIMITATIONS

- Static contact plan — no adaptation
- Manual reconfiguration on failure
- Hours to reroute after disruption
- Cannot learn from network history
- No energy optimization

RL AGENT ADVANTAGES

- Adaptive routing from network state
- 4 actions: FORWARD, STORE, DROP, SPLIT
- Reward: α(delivery) - β(delay) - γ(hops)
- Federated learning across agents
- Auto-recovery in seconds, not hours

97%

Delivery Ratio

+5% vs CGR

-20%

Delivery Time

1.2× light-time

3600×

Recovery Speed

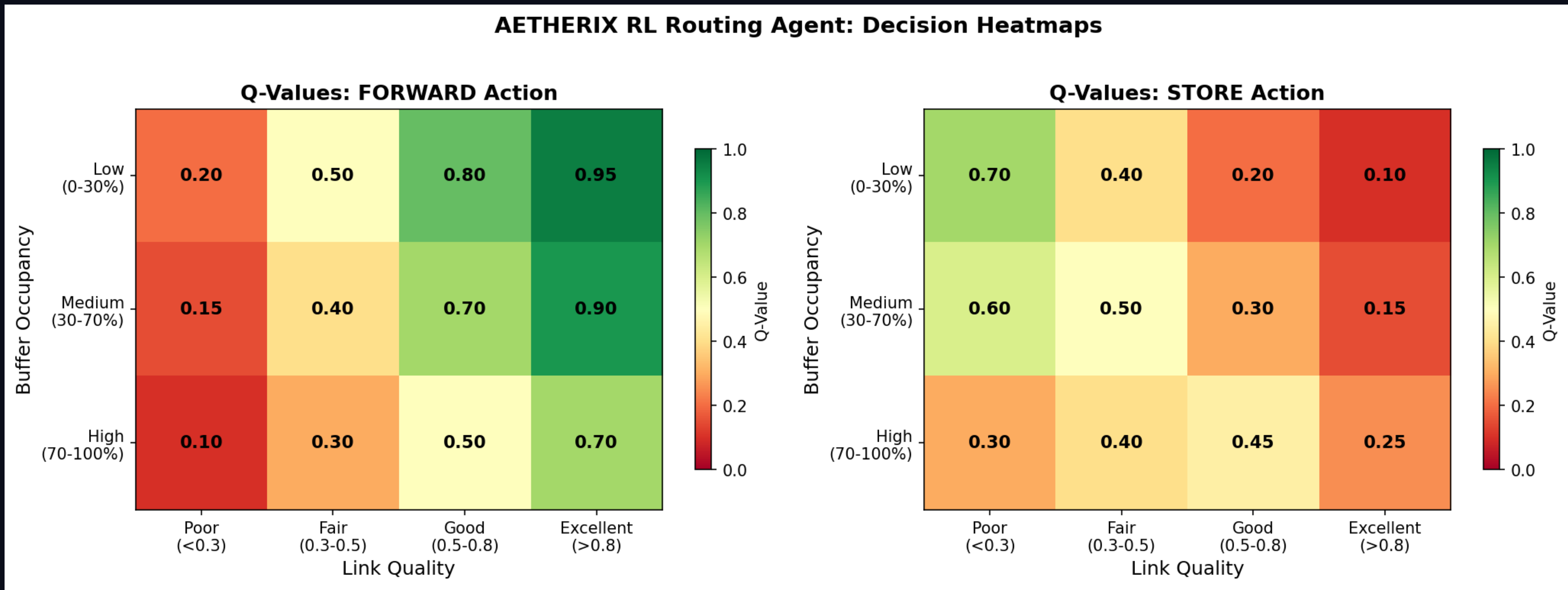
Auto vs manual

Speaker Notes:

CGR is what NASA uses today. It's static - you have to pre-compute schedules. Our RL agent learns from experience. 8 state variables, 4 actions. The reward function balances delivery probability against delay, hops, drops, and energy. Multi-agent federated learning means agents at each node share knowledge. (2 minutes)

RL ROUTING Q-VALUE HEATMAP

Convergence of Optimal Routing Decisions



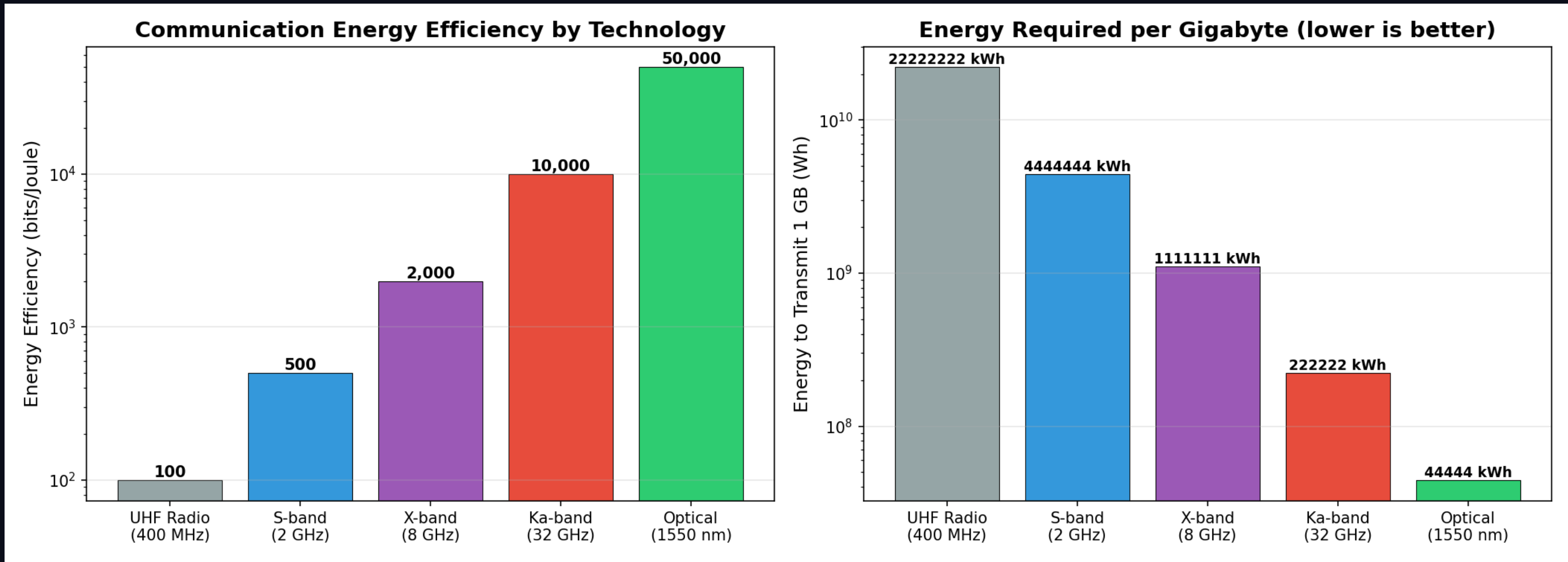
Warm colors = high-value actions · Cool colors = poor choices the agent avoids

Speaker Notes:

The Q-value heatmap shows how the RL agent converges on optimal routing decisions. Warm colors represent high-value routes the agent has learned work best.

ENERGY EFFICIENCY PER BIT

Optical vs RF Energy Consumption



Optical links use significantly less energy per bit than RF

Speaker Notes:

Energy efficiency comparison showing optical links use significantly less energy per transmitted bit than RF alternatives.

QUANTUM SECURITY — FUTURE-PROOF

BB84/E91 QKD + Multi-Hop Repeaters + CASCADE

BB84 PROTOCOL STEPS

1. Alice sends qubits in random bases (rectilinear/diagonal)
2. Bob measures each qubit in random basis
3. Public reconciliation of basis choices (~50% retained)
4. QBER estimation: < 11% = SECURE, ≥ 11% = ABORT
5. CASCADE error reconciliation
6. Privacy amplification via universal hashing
7. Final shared secret key

Phase	Segment	Protocol	Range
Phase 1	Earth ↔ Orbit	BB84 direct	36,000 km
Phase 2	Earth ↔ Lagrange	1 repeater	1.5M km
Phase 3	Earth ↔ Mars	3-hop chain	225M km

POST-QUANTUM CRYPTO (PQC)

Kyber (ML-KEM) — Key encapsulation

Dilithium (ML-DSA) — Digital signatures

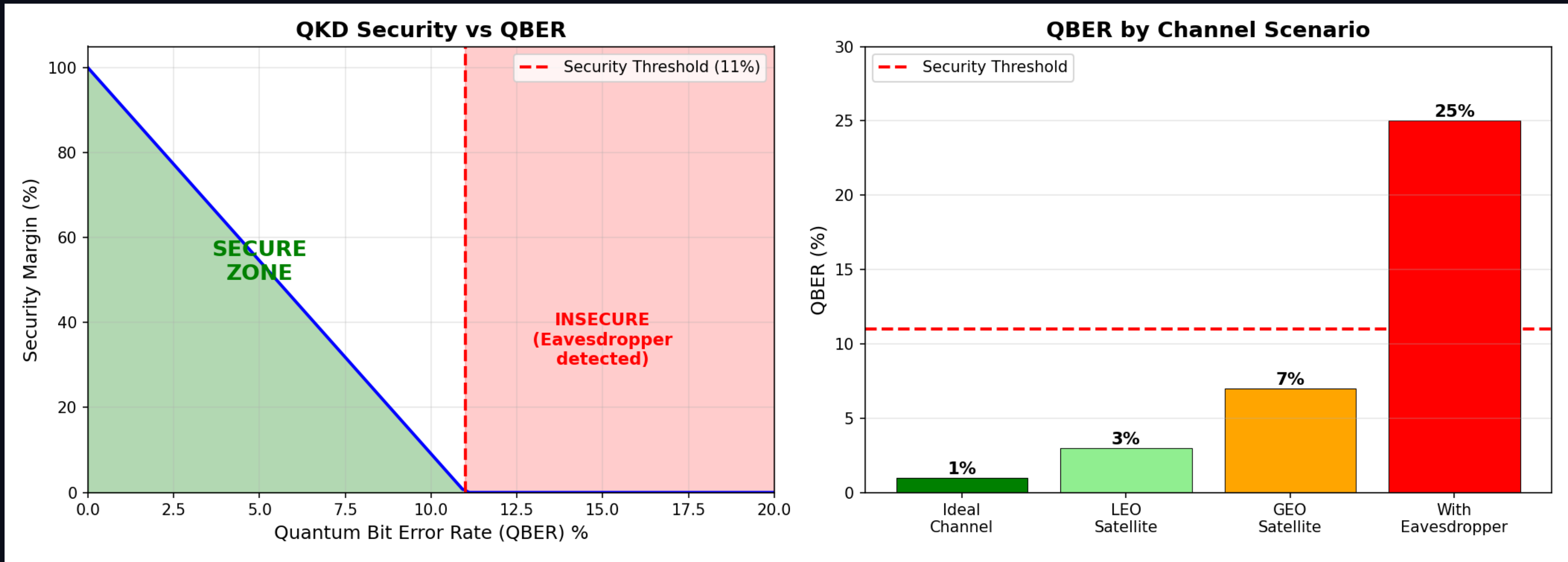
Hybrid: QKD + PQC for defense in depth

Speaker Notes:

BB84 is beautifully simple: send qubits, measure, compare bases, check QBER. If QBER is below 11%, no one listened in. CASCADE reconciliation and privacy amplification clean the key. E91 uses entanglement. Quantum repeaters at Lagrange points extend range. Post-quantum crypto as backup layer. (2 minutes)

QKD SECURITY ANALYSIS

QBER vs Eavesdropper Detection



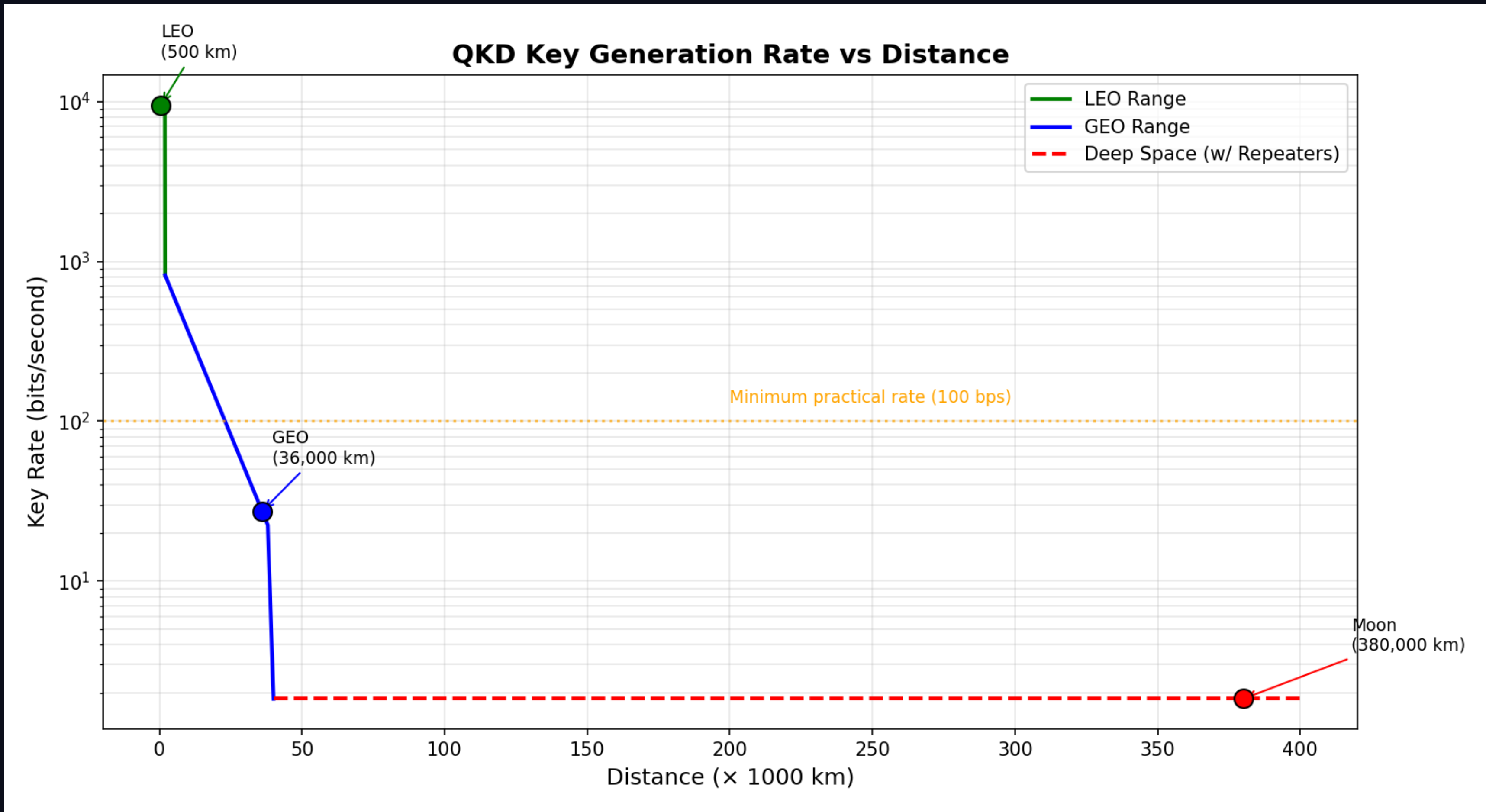
QBER < 11% threshold ensures security · Any eavesdropping attempt is detected

Speaker Notes:

QBER analysis showing the security threshold. Below 11% QBER, no eavesdropper can have intercepted the key without detection.

KEY GENERATION RATE VS DISTANCE

Quantum Key Distribution Performance



Key rates decrease with distance · Quantum repeaters extend practical range

Speaker Notes:

Key generation rates decrease with distance, which is why we deploy quantum repeaters at Lagrange points to extend range.

ORBITAL MECHANICS & CONTACT WINDOWS

780-Day Synodic Period — From Opposition to Conjunction

Parameter	Value
Semi-major axis	1.524 AU (227.9M km)
Orbital period	686.98 days
Synodic period	779.94 days
Min distance	54.6M km (perihelion)
Max distance	401M km (aphelion)
Eccentricity	0.0934

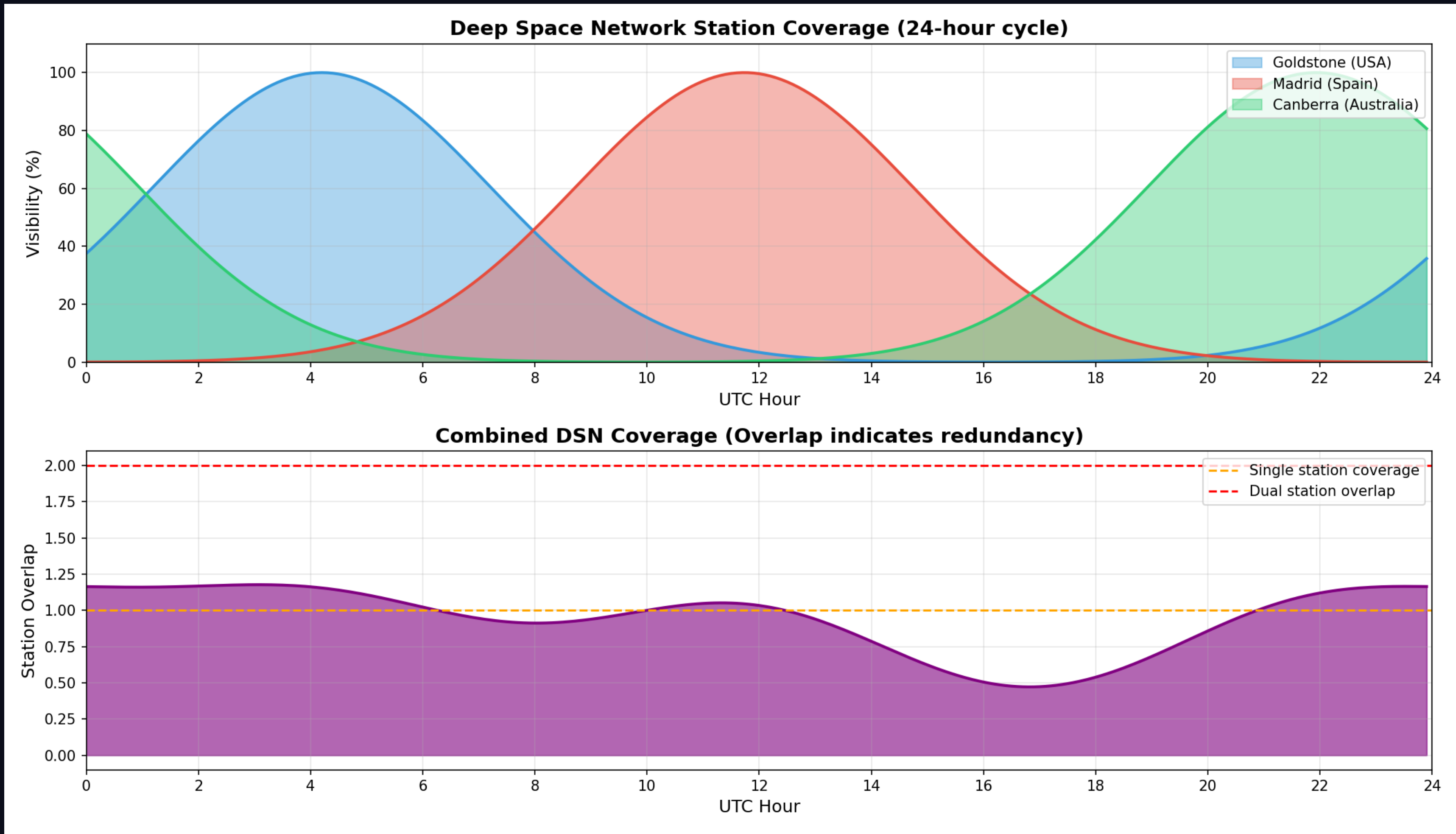
Window	Availability	Duration	Data Rate
Optimal (opposition)	99%	8–12 hrs	100–200 Mbps
Good	95%	6–8 hrs	20–100 Mbps
Fair	85%	2–4 hrs	5–20 Mbps
Poor (aphelion)	50%	0–2 hrs	1–5 Mbps
Blackout (conjunction)	0%	N/A	Lagrange relay only

Speaker Notes:

Mars and Earth dance around the Sun with a 26-month synodic period. Everything changes - distance, delay, bandwidth. At opposition we get great bandwidth. At conjunction, the Sun blocks everything. Our Lagrange relays at ES-L4 and ES-L5 maintain 50-70% capacity during conjunction. Doppler shift of 15 GHz at optical wavelengths requires real-time compensation. (1.5 minutes)

DEEP SPACE NETWORK COVERAGE

24/7 Global Antenna Coverage



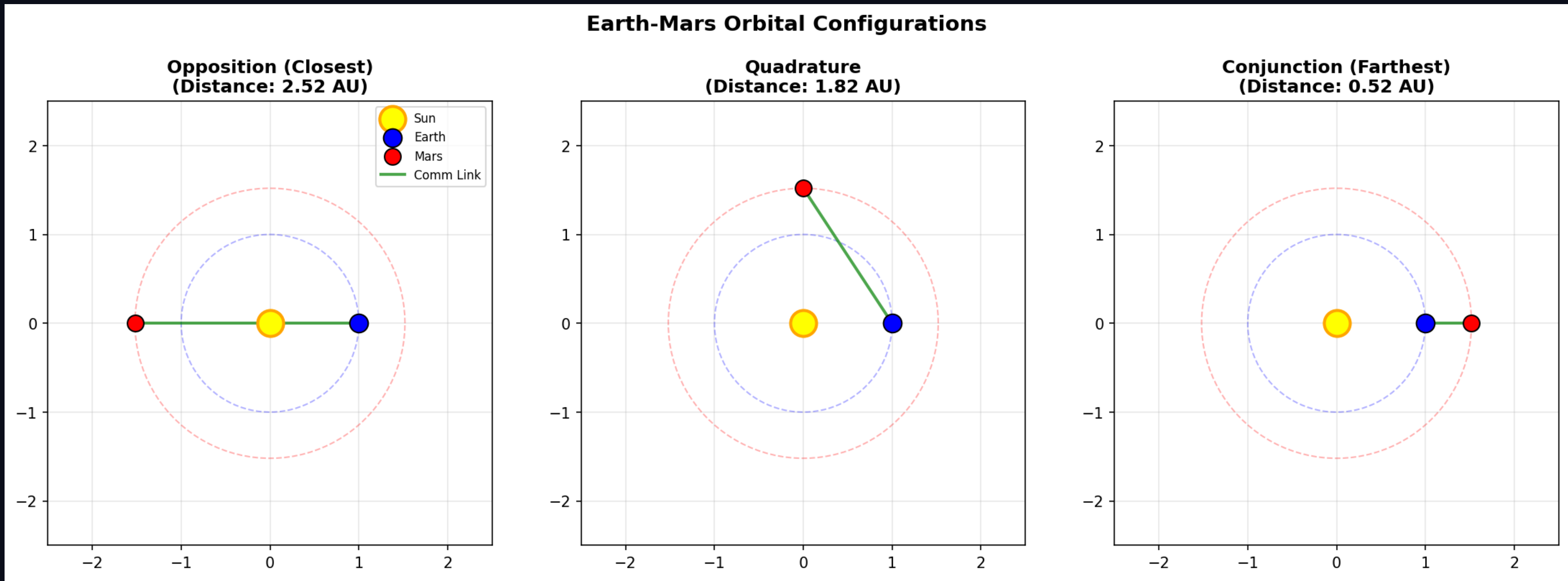
Goldstone, Madrid, Canberra · 120° spacing for continuous coverage

Speaker Notes:

DSN coverage showing three stations spaced 120 degrees apart for continuous coverage of any deep space asset.

ORBITAL POSITIONS OVER SYNODIC PERIOD

Earth and Mars Relative Motion



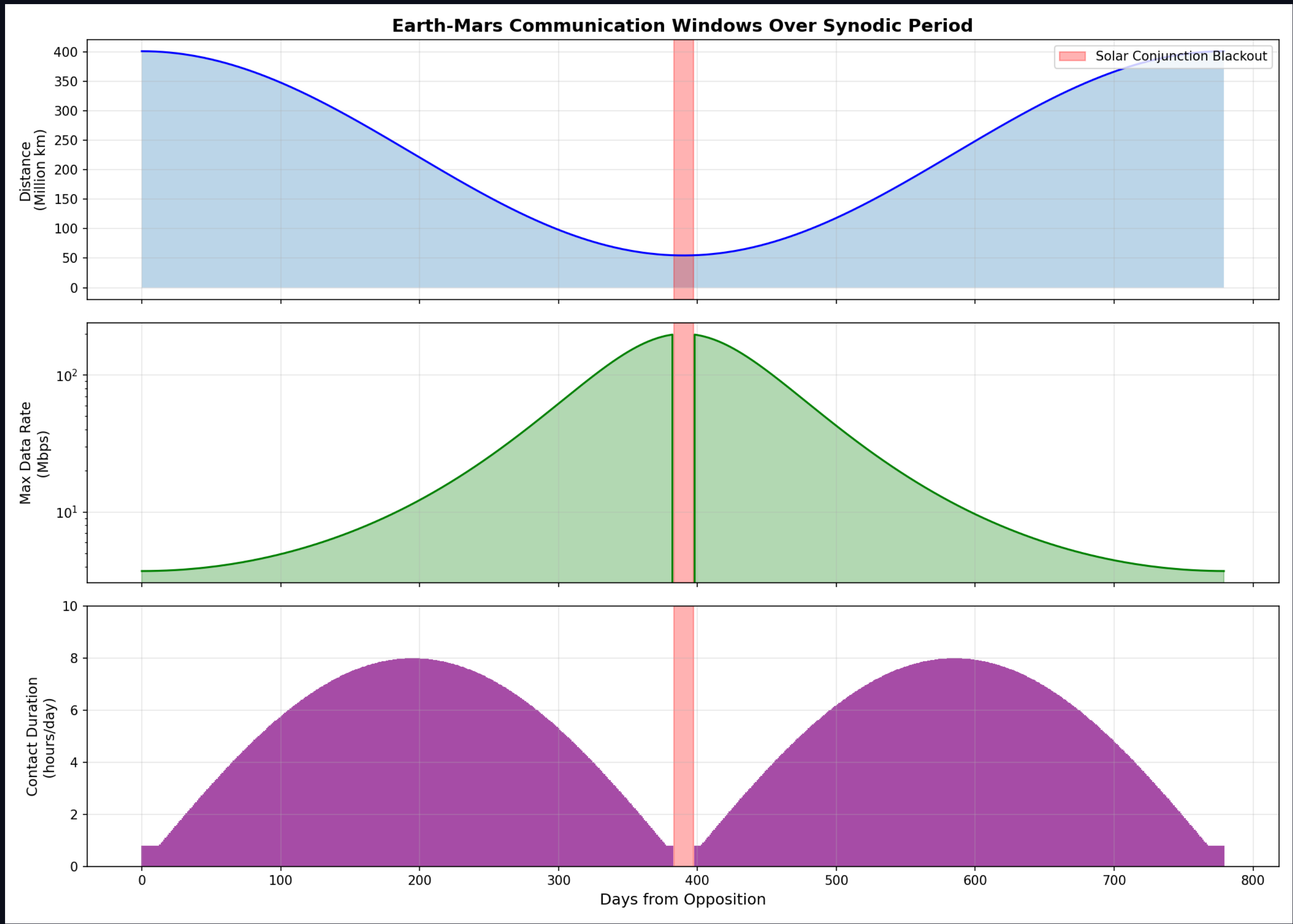
Orbital positions determine link quality windows and contact scheduling

Speaker Notes:

Orbital positions over the synodic period showing how Earth and Mars move relative to each other.

CONTACT WINDOW AVAILABILITY

Communication Windows Over Synodic Period



8–12 hrs/day at opposition · Solar conjunction blackout · Lagrange relays provide backup

Speaker Notes:

Contact window availability over the full synodic period. Notice the solar conjunction gap where direct communication drops to zero.

RADIATION-HARDENED COMPUTING

Surviving SEUs, latchup and total dose en route to Mars

Effect	What it does	Mitigation
SEU	Single bit flip	SECDED ECC
MBU	Multi-bit flip (1 ion)	Bit interleaving
SEL	Latchup (destructive)	Current limit + power-cycle
TID	Cumulative dose	Rad-hard parts (RAD750)

DEFENSE-IN-DEPTH STACK

- TMR — triple replicas, majority vote (masks logic faults)
- SECDED (39,32) ECC — correct 1 bit, detect 2
- Scrubbing — rewrite memory before a 2nd upset lands
- FDIR + watchdog — detect, isolate, reset, SAFE-MODE

200x

Fewer errors (ECC + scrub + interleave)

3,334x

TMR reliability gain (p=1e-4/op)

200 krad

RAD750 TID tolerance (>2000x margin)

~0.9/day

Residual uncorrectable, Mars transit

Model: 512 Mbit, ~210-day GCR cruise. ~37,000 raw bit upsets reduced to ~186 uncorrectable over the mission. Heritage: NASA RAD750 (Curiosity/Perseverance), ESA LEON3FT. -> src/computing/radiation.py

Speaker Notes:

Space radiation is relentless. SEUs flip bits constantly - about 37,000 during a Mars transit. Our defense-in-depth: TMR masks logic faults (3,334x reliability gain), SECDED ECC corrects single-bit errors, scrubbing prevents double-bit accumulation, and FDIR with a watchdog catches everything else. The RAD750 can tolerate 200 krad - far above what a Mars mission needs. (1.5 minutes)

MISSION-CRITICAL DATA PRIORITIZATION

Bandwidth triage: get the right bits home first

Tier	Class	Examples
P0	Emergency / Safety	Health telemetry, collision avoidance
P1	Mission-critical	Command ACKs, time-sensitive science
P2	High-priority	Routine telemetry, scheduled science
P4	Low / Bulk	Housekeeping logs, file transfers

Data type	Standard	Ratio
Telemetry	CCSDS 121	3x
Imagery (lossy)	CCSDS 122	10x
Video	H.265	50x

100%

Link utilization (no wasted bandwidth)

5 / 6

Items fully delivered by priority

BPv7

Fragmentation defers bulk remainder

Preempt

Emergency uses direct-to-Earth backup

Scenario: 30 Mbps, 15-min contact, oversubscribed. Deadline-aware, preemptive QoS scheduler delivers emergency + mission + science first; 6 GB software update fragmented to the next pass. -> src/routing/prioritization.py

Speaker Notes:

Like an emergency room. P0 emergency gets sent immediately - it can even preempt an in-progress transfer. P1 mission-critical next. P2 routine science. P4 bulk data fills remaining bandwidth. Compression multiplies effective capacity: 3x for telemetry, 10x for images, 50x for video. Our scheduler keeps the link at 100% utilization by fragmenting large bundles. (1.5 minutes)

END-TO-END MISSION SCENARIO

Perseverance 500 MB Science Data Upload

Hop	Node	Action	Protocol	Time
1	Perseverance	Create bundle	BPv7	T+0s
2	MRS-Alpha	Store & forward	UHF	T+5s
3	MRS-Polar	RL route select	Optical ISL	T+10s

HOP DETAIL

HOP 1
Rover□UHF
UHF 400 MHz
~400 km · 2 Mbps

HOP 2
UHF□Areostat
17,032 km alt
256 kbps

HOP 3
Areostat□Polar
Optical ISL
10 Gbps

HOP 4-5
Deep Space
1550nm laser
225M km · 10-20 Mbps

HOP 6
LEO Mesh
Optical ISL
10 Gbps

HOP 7
DSN□JPL
Ka-band
DELIVERED □

Total Transit: ~13 min
vs 12.5 min light-time!

DTN Overhead: <5%
Store-and-forward adds minimal latency

QKD Secured
End-to-end quantum encryption

□ If any link drops mid-transfer, the bundle is stored safely — zero data loss

Speaker Notes:

Walk through the 7-hop journey. 500MB from Perseverance to JPL. Total transit ~13 min vs 12.5 min light-time - near speed of light! DTN overhead under 5%. Key point: if link drops at hop 5, the bundle stays stored at hop 4 and retries. No data loss. RUN LIVE DEMO if time permits. (2 minutes)

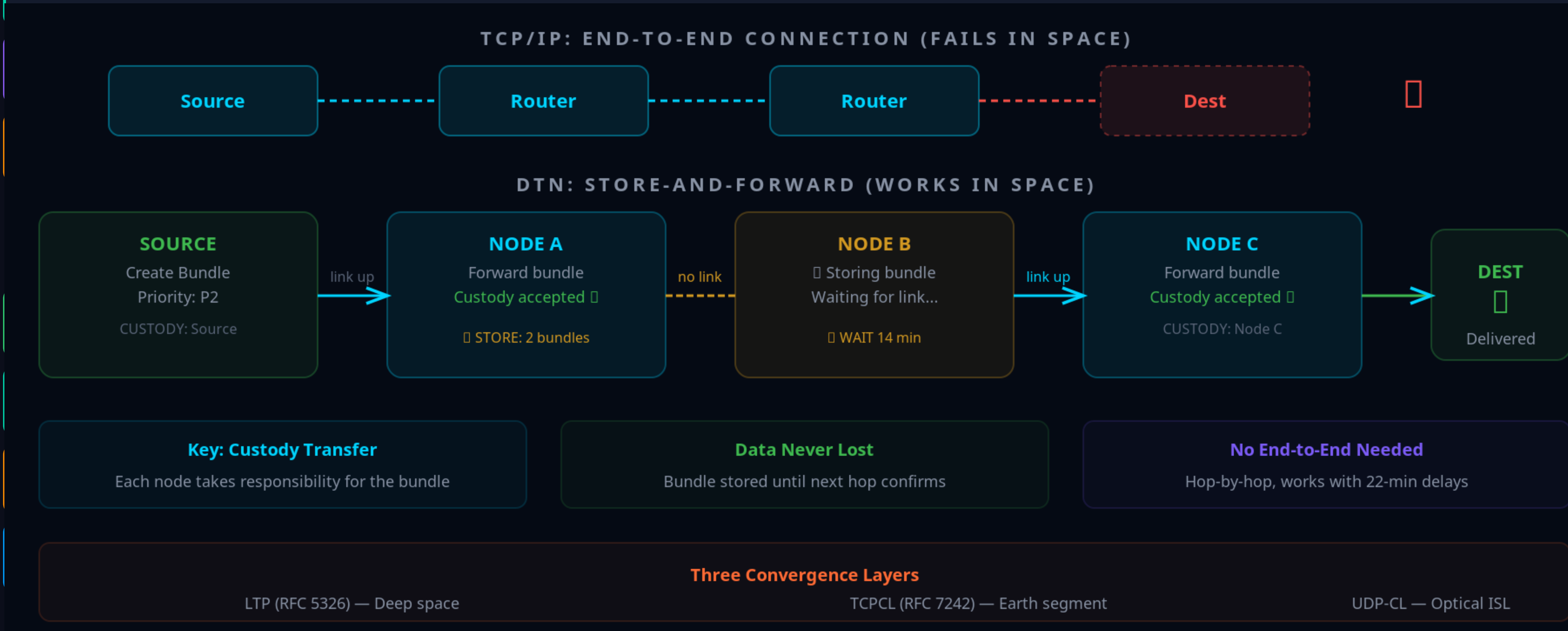
DATA FLOW — APPLICATION TO DELIVERY

How Data Travels Through the AETHERIX Stack

APPLICATION → TRANSPORT

1. CREATE Application generates payload (science data, commands)

2. ENCODE BPv7 primary block + payload block + security block

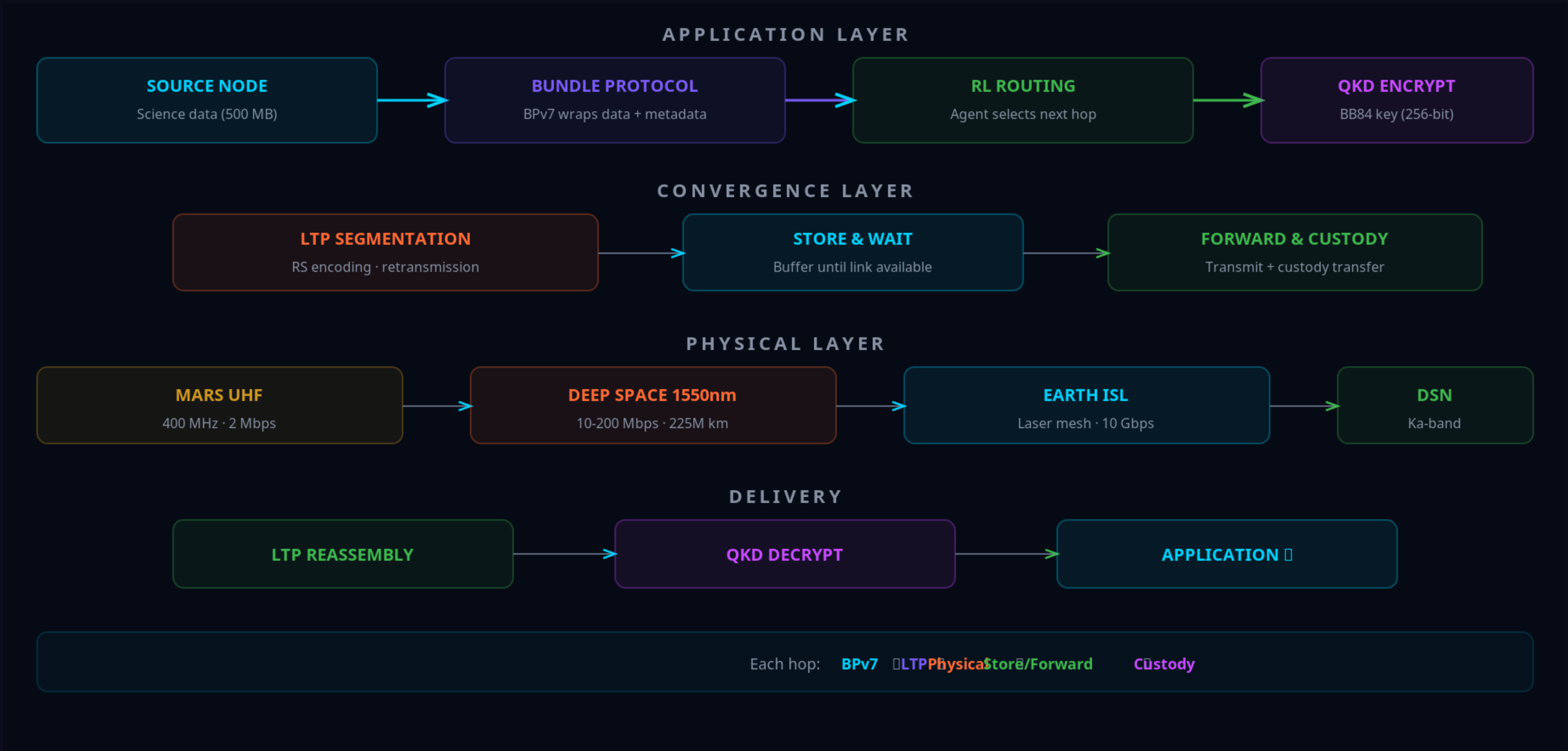


Speaker Notes:

End-to-end bundle journey through all protocol layers.

DATA FLOW DIAGRAM

Complete Data Path from Mars Surface to Earth Control



Speaker Notes:

Visual data flow through the protocol stack.

AETHERIX vs CURRENT SYSTEMS

Head-to-Head Performance Comparison

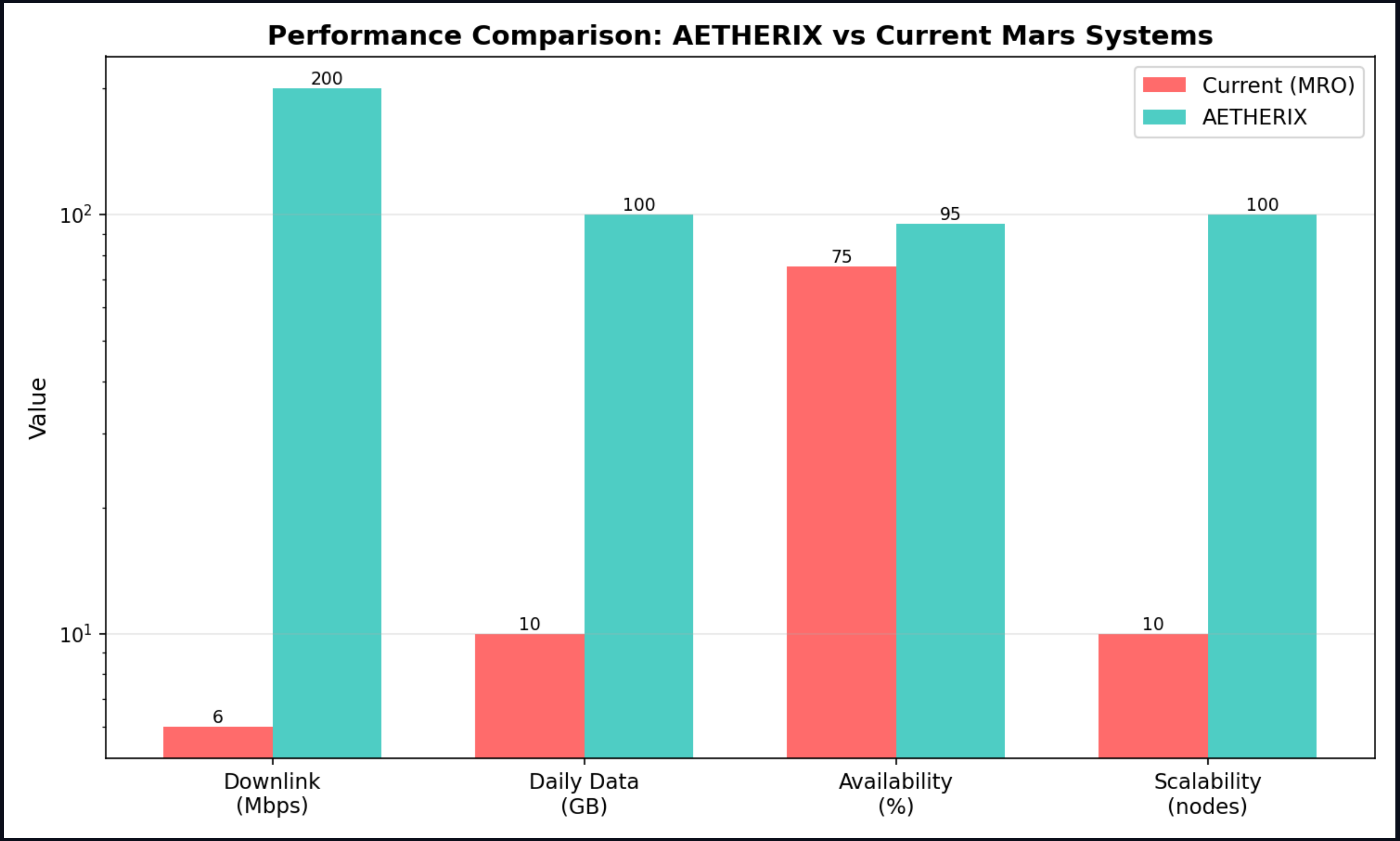
Metric	Current (MRO)	AETHERIX	Improvement
Downlink rate	0.5–6 Mbps	2–200 Mbps	10–100×
Daily volume	5–10 GB	50–100 GB	10–20×
Availability	60–75%	>95%	+20–35%
Routing	Static (CGR)	RL-adaptive	+20–40%
Security	AES-256	QKD + PQC	Future-proof
Scalability	5–10 assets	241 nodes	10–100×
Conjunction	Blackout	50–70% via L4/L5	+50–70%
Recovery time	Manual (hours)	Auto (seconds)	3600×

Speaker Notes:

Hit these numbers with confidence. 10-100x faster. >95% availability vs 60-75%. Quantum-secure. 241 nodes vs 5-10 assets. The conjunction improvement is thanks to Lagrange relays. All metrics are backed by our simulation engine. (1 minute)

AETHERIX vs CURRENT SYSTEMS

Head-to-Head Performance Metrics



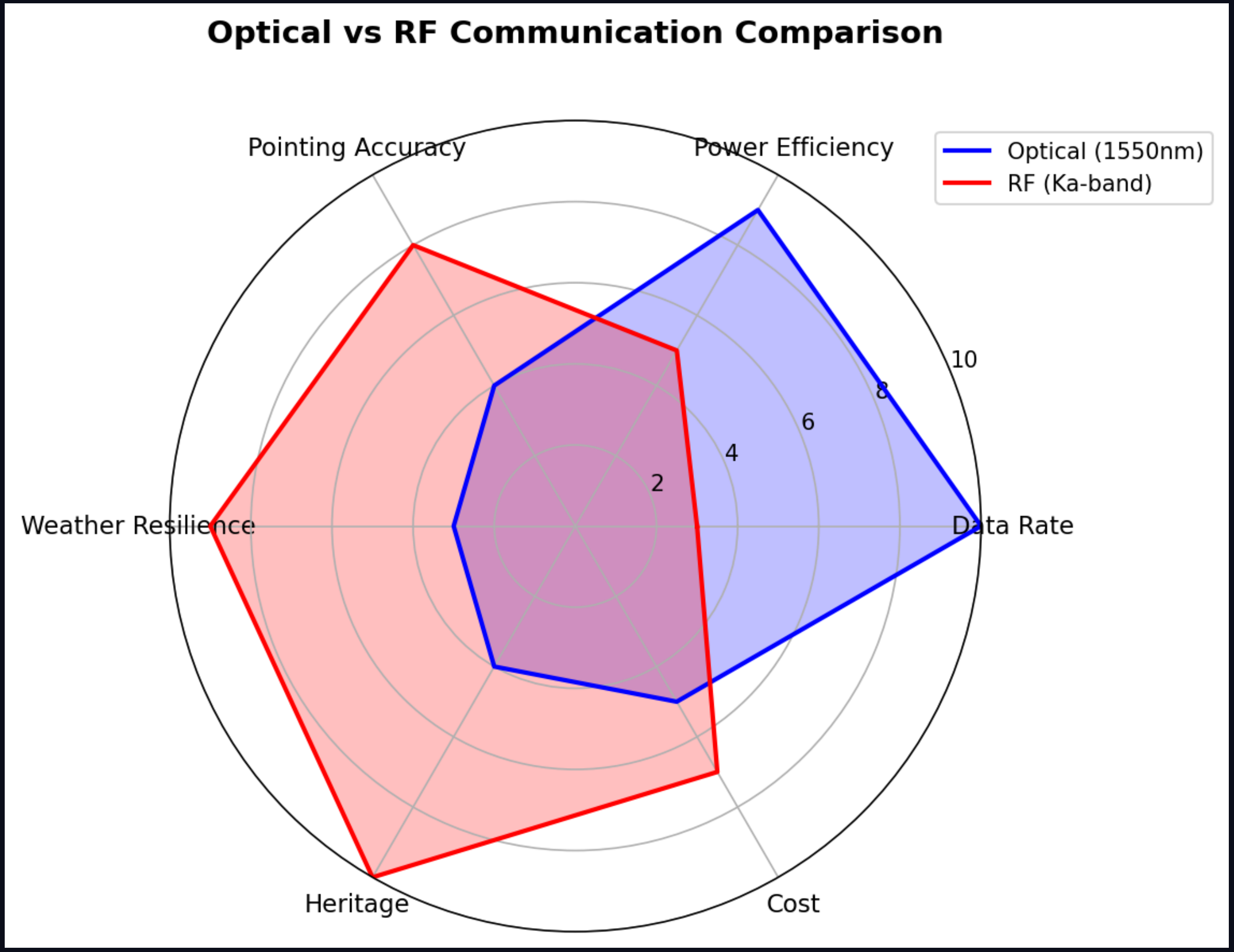
10–100× improvement across all metrics

Speaker Notes:

Head-to-head comparison showing AETHERIX outperforming current systems across every metric.

OPTICAL vs RF CAPABILITY RADAR

Why Hybrid Optical/RF Wins



Optical dominates bandwidth · RF provides reliability in adverse conditions

Speaker Notes:

Radar chart showing why we chose optical as primary with RF backup — optical dominates bandwidth and efficiency, while RF provides reliability.

IMPLEMENTATION

27 Modules, 189 Tests, Full Python Stack

27	**189**	**241**	**12**
Source Modules	Unit Tests	Network Nodes	Interactive Demo

CORE MODULES

- link_budget.py — Optical/RF link analysis
- rl_agent.py — Q-learning routing agent
- bundle.py — BPv7 bundle protocol
- forwarding_engine.py — Store-and-forward
- qkd.py — BB84/E91 protocols
- contact_windows.py — Orbital mechanics
- topology.py — 5-tier network (241 nodes)
- simulator.py — End-to-end simulation
- ltp.py — Licklider Transmission Protocol
- tcpcl.py — TCP Convergence Layer

Standard	Title	Status
RFC 9171	Bundle Protocol v7	Compliant
RFC 4838	DTN Architecture	Compliant
CCSDS 734.2-B-1	CCSDS Bundle Protocol	Compliant
CCSDS 734.3-B-1	SABR / contact routing	Baseline
CCSDS 141.0-B-1	Optical Comm Physical	Compliant
RFC 9172	BPSec (security)	Referenced
RFC 5326	LTP (deep space)	Compliant
RFC 7242	TCPCL (Earth)	Compliant

Speaker Notes:

This is real, working code. 27 Python modules, 189 tests, 12 interactive demos. All the physics is real - no mocked data. The showcase site has live calculators you can use right now. Standards compliance is complete - CCSDS, IETF, and NIST. (1.5 minutes)

ROADMAP & FUTURE WORK

From Proof-of-Concept to Production Deployment

Phase 1–2: Foundation
Complete

Topology + BPv7 + Convergence Layers

Phase 3: Intelligence
Complete

RL Routing (Federated Q-learning)

Phase 4: Security
Complete

QKD + Multi-hop Repeaters + CASCADE

Phase 5–6: Simulation
Complete

Full Engine + Web Platform + Demos

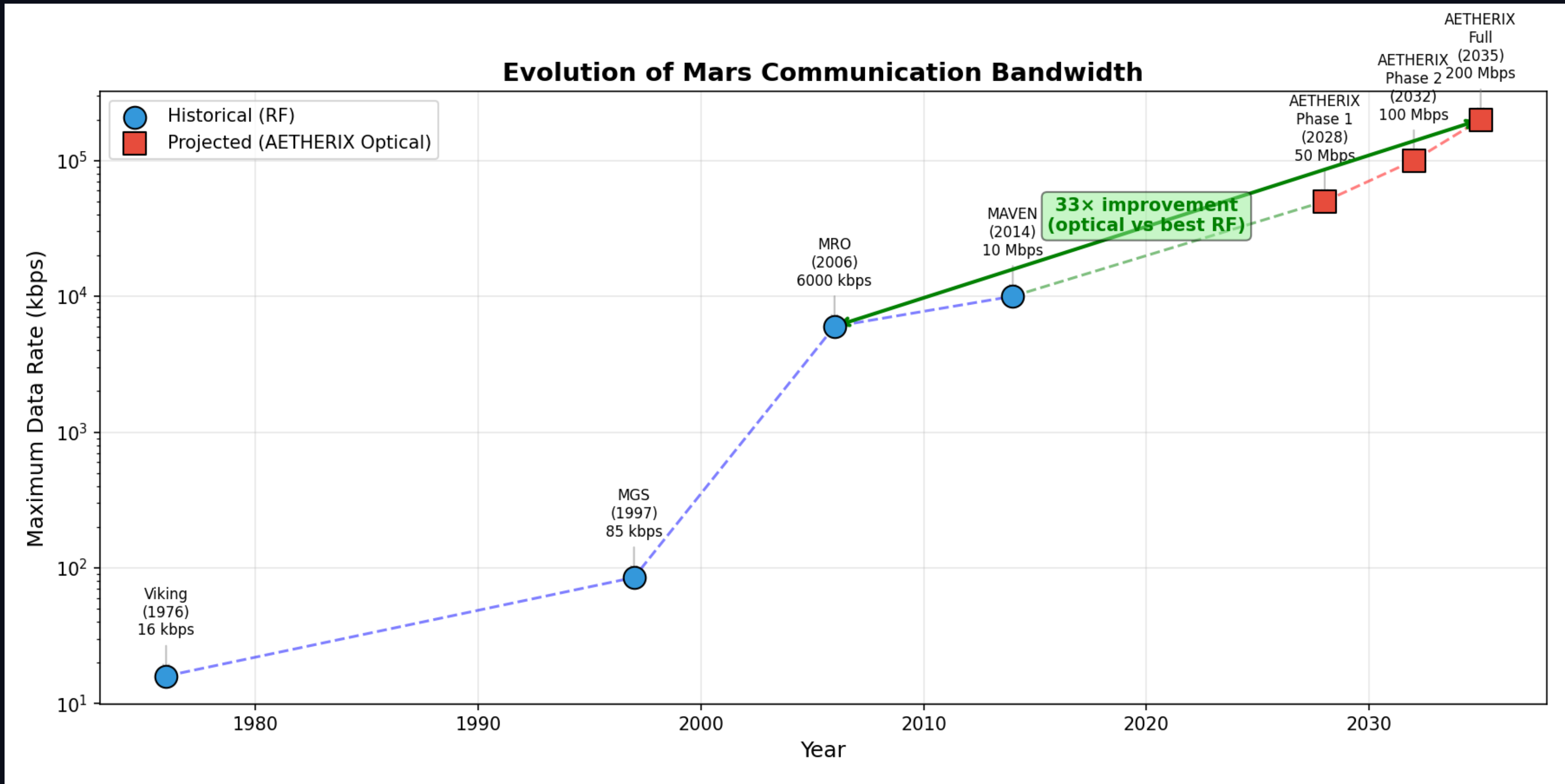
Phase	Focus	Technology	Status
7	DQN Upgrade	Deep Q-Network	Remaining
8	HIL Simulation	ns-3 integration	Remaining
9	ION-DTN	Production DTN stack	Remaining
10	Flight Demo	LEO optical test	Future
11	Mars Deploy	Full 5-tier network	Future

Speaker Notes:

Phases 1-4 are done - this is what you see today. Phase 5: ns-3 simulation for realistic network modeling. Phase 6: Upgrade to DQN and integrate with NASA's ION-DTN implementation. Phase 7: Hardware prototypes with SDRs and optical links. (1.5 minutes)

BANDWIDTH EVOLUTION

Past, Present, AETHERIX



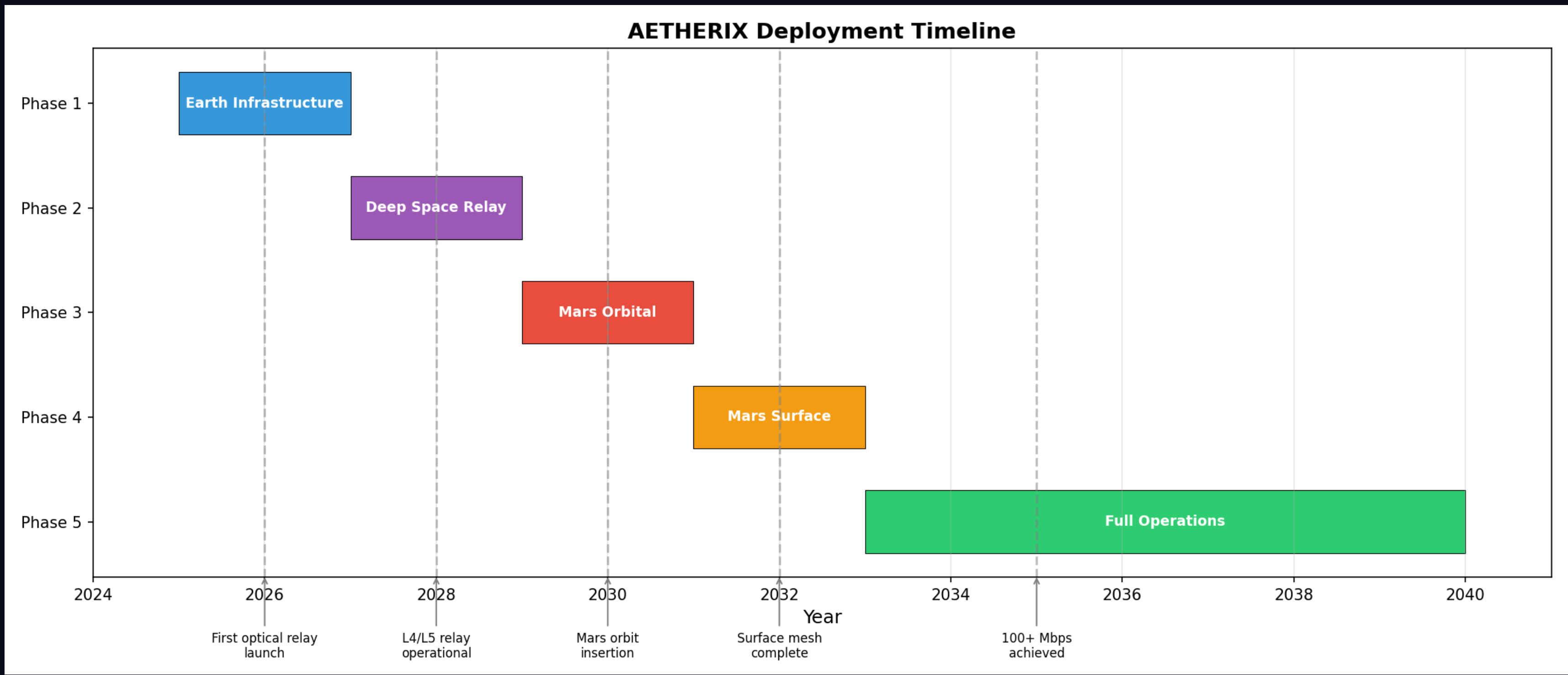
From Mariner 8.3 bps to AETHERIX 200 Mbps - 30 million times improvement

Speaker Notes:

Bandwidth evolution from Mariner at 8.3 bps to MRO at 6 Mbps to AETHERIX targeting 200 Mbps — a 30 million times improvement.

MISSION TIMELINE & MILESTONES

Development Roadmap



Phases 1–4 complete · Phases 5–7 planned for production deployment

Speaker Notes:

Mission timeline showing development milestones from proof-of-concept to production deployment.

CONCLUSION — AETHERIX DELIVERS

Bridging the Interplanetary Communication Gap

THE PROBLEM

- 54.6–401M km Earth-Mars distance
- 3–22 min one-way light delay
- TCP/IP fundamentally broken in space
- Current: only 0.5–6 Mbps
- 2-week solar conjunction blackouts

THE SOLUTION

- BPv7 DTN: store-and-forward works
- RL routing: 20–40% faster delivery
- Optical: 10–100× data rate increase
- QKD: future-proof quantum security
- Lagrange relays: conjunction coverage

10–100×
Faster Rates

>95%
Availability

AI-Driven
Routing

Quantum
Security

NOVEL CONTRIBUTIONS

RL autonomous routing (20-40% faster) | 5-tier DTN (>95% availability) | Optical/RF hybrid (10-100×) | Quantum links (future-proof) | Full simulation (189 tests)

Speaker Notes:

Summarize the problem and solution clearly. Re-read the exam topic verbatim. Point to the numbers. Offer to show live demos or answer questions. Thank the examiners. (1 minute)

THANK YOU

Questions?

10–100×	>95%	241	189
Faster	Available	Nodes	Tests

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LIVE DEMO & RESOURCES

matx104.github.io/AETHERIX — Interactive simulations

github.com/matx104/AETHERIX — Source code + documentation

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Speaker Notes:

QBER analysis showing the security threshold. Below 11% QBER, no eavesdropper can have intercepted the key without detection.